

Let's consider $SU(2)$ case first from Bijnens 1

$$\mathcal{L} = \bar{q}_L i \gamma^\mu D_\mu q_L + \bar{q}_R i \gamma^\mu D_\mu q_R - \bar{q}_R M_{ij} q_{Lj} - \bar{q}_{Li} M_{ij}^+ q_{Rj}$$

$$D_\mu q = \partial_\mu q - i G_\mu q$$

$$G_\mu = G_\mu^a \sigma^a$$

(here we have
2 N_F U(1) d.o.f.s)

$$q \rightarrow e^{i d^a \sigma^a} q$$

- transformation of q under $SU(2)$

$$\delta_2 = i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i \varepsilon, \quad \varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \varepsilon^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\mathbb{1}$$

$$\delta_2 \vec{\delta} \delta_2 = -\vec{\delta}^T$$

We define new object: $C = i \gamma^2 \gamma^0$

$$\tilde{q}_{R\alpha,i} = \varepsilon_{\alpha\beta} C \bar{q}_{L\beta}^T$$

here α, β - group indices (color)

$$P_L \tilde{q}_R = 0, \quad P_R \tilde{q}_R = \tilde{q}_R \Rightarrow \text{right-handed}$$

How $\tilde{q}_R$ does transform?

(omit flavour index)

$$\begin{aligned} \tilde{q}_{R\alpha} &= \varepsilon_{\alpha\beta} C \bar{q}_{L\beta}^T = \varepsilon_{\alpha\beta} C (q_{L\beta}^\dagger \gamma^0)^T = \\ &= \varepsilon_{\alpha\beta} C \gamma^{0T} q_{L\beta}^* \xrightarrow{\text{gauge}} \varepsilon_{\alpha\beta} C \gamma^{0T} \left(e^{i d^a \delta^a} q_{L\beta} \right)^* = \\ &= \varepsilon_{\alpha\beta} C \gamma^{0T} \left(e^{-i d^a (\delta^a)^*} q_{L\beta}^* \right) \quad \textcircled{=} \end{aligned}$$

$$\delta_a^+ = \delta_a^* \Rightarrow \delta_a^* = \delta_a^T$$

$$\text{expanding: } \varepsilon_{\alpha\beta} (\delta_{\beta\gamma} - i d^a \delta_{\alpha,\beta\gamma}^T) q_\gamma^* \quad \textcircled{=}$$

key moment: we use that fact that we can connect δ^* with δ !

$$\text{We know: } \delta_2 \vec{\delta} \delta_2 = -\vec{\delta}^T$$

$$-i \varepsilon \vec{\delta} (-i \varepsilon) = -\vec{\delta}^T \Rightarrow \varepsilon \vec{\delta} \varepsilon = \vec{\delta}^T \quad | \cdot \varepsilon \Rightarrow -\vec{\delta} \varepsilon = \varepsilon \vec{\delta}^T$$

$$\textcircled{=} \varepsilon (1 - i d^a \delta_a^T) q^* = (1 + i d^a \delta_a) \varepsilon q^* = e^{i d^a \delta_a} (\varepsilon q)$$

$$\textcircled{=} C \gamma^0 T \cdot e^{i d^a \delta^a} (\varepsilon q) = e^{i d^a \delta^a} (C \gamma^0 T \varepsilon_{\alpha\beta} q_\beta) = e^{i d^a \delta^a} \varepsilon_{\alpha\beta} \gamma^0 T q_\beta = e^{i d^a \delta^a} \tilde{q}_R$$

$\uparrow$
C acts on dirac indices

i.e. $\tilde{q}_R$ transforms same as q_R

$\Rightarrow$ we can build a new object

$$\hat{q} = \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix}$$

$\nwarrow$ N_f R-handed Dirac spinors (i.e. N_f Weyl d.o.f.s)

$$\hat{q} \rightarrow e^{i d^a \delta^a} \hat{q} \quad (\text{transformation under } SU(2))$$

$\nearrow$ corresponds to N_f L-handed Dirac spinors (i.e. N_f Weyl d.o.f.s)

We can write new d:

$$\mathcal{L} = \bar{\hat{q}} i \gamma^\mu D_\mu \hat{q} - \frac{1}{2} \bar{\hat{q}}_\alpha C \varepsilon_{\alpha\beta} \hat{M} \hat{q}_\beta - \frac{1}{2} \hat{q}_\alpha \varepsilon_{\alpha\beta} C \hat{M}^+ \hat{q}_\beta$$

kinetic term

$$\bar{\hat{q}} i \gamma^\mu D_\mu \hat{q} = (\bar{q}_R \tilde{\bar{q}}_R) \cdot i \gamma^\mu D_\mu \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix} = (\bar{q}_R \tilde{\bar{q}}_R) \begin{pmatrix} i \gamma^\mu D_\mu q_R \\ i \gamma^\mu D_\mu \tilde{q}_R \end{pmatrix} =$$

$$= i \bar{q}_R \not{D} q_R + i \tilde{\bar{q}}_R \not{D} \tilde{q}_R$$

$$\tilde{q}_R = \varepsilon \cdot C \bar{q}_L^T \quad (C = i \gamma^2 \gamma^0)$$

$$\begin{aligned} \Rightarrow \tilde{\bar{q}}_R &= (\tilde{q}_R)^\dagger \gamma^0 = (\varepsilon \cdot C \bar{q}_L^T)^\dagger \gamma^0 = \\ &= (\bar{q}_L)^{T\dagger} C^\dagger \varepsilon^\dagger \gamma^0 = (q_L^\dagger \gamma^0)^{T\dagger} C^\dagger \varepsilon^\dagger \gamma^0 = \\ &= (\underbrace{\gamma^0^T}_{=\gamma^0} q_L^{\dagger T})^\dagger C^\dagger \underbrace{\varepsilon^\dagger}_{=-\varepsilon} \gamma^0 = -q_L^T \gamma^0 C^\dagger \varepsilon \gamma^0 \end{aligned}$$

$$\boxed{ i \tilde{\bar{q}}_R \not{D} \tilde{q}_R = -i q_L^T \gamma^0 C^\dagger \varepsilon \gamma^0 \not{D} \varepsilon C \cdot \bar{q}_L^T \quad \textcircled{=}$$

(sum over flavour is suppressed)

$$C^\dagger = (i \gamma^2 \gamma^0)^\dagger = -i \gamma^{0\dagger} \gamma^{2\dagger} = -i \gamma^0 \cdot \gamma^0 \gamma^2 \gamma^0 = -i \gamma^2 \gamma^0$$

$$\gamma^{\mu\dagger} = \gamma^0 \gamma^\mu \gamma^0$$

$$\textcircled{=} -i q_L^T \gamma^0 (\underline{-i \gamma^2 \gamma^0}) \varepsilon \gamma^0 \not{D} \varepsilon (\underline{i \gamma^2 \gamma^0}) \cdot (q_L^\dagger \gamma^0)^T =$$

If we had another theory, for example fundamental rep of $Sp(4)$ then we will have in the definition of ε instead of ε - another matrix which connects the gauge group generators with its $*$.

Condensate

$$\mathcal{L}_{\text{pseudoscalar}} = -\bar{q}_R M_{ij} q_L - \bar{q}_L M_{ij}^+ q_R$$

$$\langle \bar{q} q \rangle = \sum_{i=1}^{N_f} \langle \bar{q}_i q_i \rangle = \sum_{i=1}^{N_f} \langle \bar{q}_L q_R + \bar{q}_R q_L \rangle$$

Note

$$\begin{aligned} \bar{\psi} \psi &= \psi^\dagger \gamma^0 \psi = \psi^\dagger \gamma^0 (P_L \psi + P_R \psi) = (P_L \psi + P_R \psi)^\dagger \gamma^0 (P_L \psi + P_R \psi) = \\ &= (\psi^\dagger P_L^\dagger + \psi^\dagger P_R^\dagger) \gamma^0 (P_L \psi + P_R \psi) = (\psi^\dagger P_L + \psi^\dagger P_R) \gamma^0 (P_L \psi + P_R \psi) = \{ \gamma^0, \gamma^5 \} = 0 \\ &= (\psi^\dagger P_L + \psi^\dagger P_R) \underbrace{\gamma^0}_{\neq 0} (P_R \psi + P_L \psi) = \psi^\dagger P_L P_L \gamma^0 \psi + \psi^\dagger P_R P_R \gamma^0 \psi = \\ &= \psi^\dagger P_L \gamma^0 P_R \psi + \psi^\dagger P_R \gamma^0 P_L \psi = \psi_L^\dagger \gamma^0 \psi_R + \psi_R^\dagger \gamma^0 \psi_L = \bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L \end{aligned}$$

$\langle \bar{q} q \rangle$ can be rewritten using $\hat{q}$ & $J_A = \begin{pmatrix} 0 & -1_{N_f} \\ 1_{N_f} & 0 \end{pmatrix}$ as

$$\langle \hat{q} \varepsilon C J_A \hat{q} \rangle + \text{h.c.}$$

group Dirac flavour

Where is the phase?!

$$\tilde{q}_R = \varepsilon C \bar{q}_L^T, \quad C = i\gamma_2 \gamma_0$$

Check the second term of $\mathcal{L}_{\text{mass}}$

$$-\frac{1}{2} \left[\bar{q} C \exp \hat{A} \hat{q}^T \right]^\dagger = -\frac{1}{2} \tilde{q}^* M^\dagger C_P^\dagger C^\dagger \tilde{q}^\dagger \quad \odot$$

$$C^+ = -i(\gamma^1 \gamma^0)^+ = -i \gamma^0 \gamma^{1+} = -i \underbrace{\gamma^0 \gamma^0 \gamma^1 \gamma^0}_1 = -i \gamma^1 \gamma^0 = -C^+$$

$$\varepsilon^+ = -\varepsilon$$

$$\ominus \frac{-1}{2} (\bar{q}^+ \gamma^0)^* M^+ \varepsilon_{\alpha\beta} C (q^+ \gamma^0)^+ = -\frac{1}{2} \bar{q}^+ \gamma^0 M^+ \varepsilon_{\alpha\beta} C \gamma^0 q^+ = +\frac{1}{2} \bar{q}^+ M^+ \varepsilon C q^+ = -C$$

$$\gamma^{0+} = \gamma^0$$

$$(C^+ C^+ = \gamma^0 \gamma^2 \gamma^0 \gamma^0 = -i \gamma^2 \gamma^0 \gamma^0 \gamma^0 = -i \gamma^2 \gamma^0 = -C)$$

∴ second term is

$$\frac{1}{2} \bar{q}^+ \varepsilon C M^+ q^+ = \frac{1}{2} (\bar{q}_R^T, \bar{q}_L^T) \varepsilon C M^+ \begin{pmatrix} q_R \\ q_L \end{pmatrix} =$$

$$= \frac{1}{2} (\bar{q}_R^T, \bar{q}_L^T C^T \varepsilon^T) \varepsilon C \begin{pmatrix} 0 & M^* \\ -M^+ & 0 \end{pmatrix} \begin{pmatrix} q_R \\ \varepsilon C \bar{q}_L^T \end{pmatrix} =$$

$$= \frac{1}{2} (\bar{q}_R^T, \bar{q}_L^T C^T \varepsilon^T) \varepsilon C \begin{pmatrix} M^* \varepsilon C \bar{q}_L^T \\ -M^+ q_R \end{pmatrix} =$$

$$= \frac{1}{2} (\bar{q}_R^T \varepsilon C M^* \varepsilon C \bar{q}_L^T - \bar{q}_L^T C^T \varepsilon^T \cdot \varepsilon C M^+ q_R) \ominus$$

$$\varepsilon^2 = -1$$

$$\varepsilon^T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -\varepsilon \Rightarrow \varepsilon^T \varepsilon = -\varepsilon \varepsilon = -\varepsilon^2 = +1$$

$$C^2 = i \gamma^1 \gamma^0 i \gamma^1 \gamma^0 = - \gamma^2 \gamma^0 \gamma^2 \gamma^0 = + \underbrace{\gamma^2 \gamma^2}_{-1} \underbrace{\gamma^0 \gamma^0}_1 = -1$$

$$C^T = i(\gamma^2 \gamma^0)^T = i \gamma^{0T} \gamma^{2T} = i \gamma^0 \gamma^{2T} = i \gamma^0 \gamma^2 = -i \gamma^1 \gamma^0 = -C$$

$$\Rightarrow C^T C = \underbrace{-C \cdot C}_{-1} = 1$$

$$\ominus \frac{1}{2} (\bar{q}_R^T M^* \bar{q}_L^T - \bar{q}_L^T M^+ q_R) = \frac{1}{2} (-\bar{q}_L^T M^+ q_R - \bar{q}_L^T M^+ q_R) = -\bar{q}_L^T M^+ q_R$$

$$\left\{ \begin{aligned} \bar{q}_R^T M^* \bar{q}_L^T &= \bar{q}_R^T M^* (\bar{q}_L^+ \gamma^0)^T = \bar{q}_R^T M^* \gamma^{0T} q_L^+ = q_R^{\alpha, M} M^{* \mu \nu} \gamma^{0 \beta} q_L^{* \beta, \mu} = \\ &= - q_L^{* \beta, \mu} \gamma^{0 \gamma \alpha} (M^*)^{\mu \nu} q_R^{\alpha, \nu} = - q_L^+ \gamma^0 M^+ q_R = -\bar{q}_L^T M^+ q_R \end{aligned} \right\}$$

$\Rightarrow$ the correct mass term:

$$\mathcal{L}_{uv} \supset \underbrace{-\bar{q}_R \mathcal{M} q_L - \bar{q}_L \mathcal{M}^\dagger q_R}_{\text{orange bracket}} = \underbrace{-\frac{1}{2} \hat{q}^T C_\epsilon \hat{u} \hat{q}^{\bar{T}} + \frac{1}{2} \hat{q}^{\bar{T}} C_\epsilon \hat{u}^\dagger \hat{q}}_{\text{orange bracket}}$$

$$\hat{u} = \begin{pmatrix} 0 & -\mathcal{M} \\ \mathcal{M}^\dagger & 0 \end{pmatrix}$$

$$\mathcal{M} = \begin{pmatrix} m_u & 0 \\ 0 & m_d \end{pmatrix}$$

$$J_A = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}, \quad I = \text{unit } N_f \times N_f$$

the condensate is

$$\langle \bar{q} q \rangle \equiv \sum_{i=1}^{N_f} \langle \bar{q}_i q_i \rangle \stackrel{\text{assume sum over flavours}}{=} \langle \bar{q}_L q_R + \bar{q}_R q_L \rangle = \langle \bar{q}_L I q_R + \bar{q}_R I q_L \rangle = \left\{ \begin{array}{l} \text{we just} \\ \text{replace} \\ \mathcal{M} \text{ by } I \end{array} \right\}$$

$$= \left\langle -\frac{1}{2} \hat{q}^T C_\epsilon \begin{pmatrix} 0 & -I \\ I^\dagger & 0 \end{pmatrix} \hat{q} + \frac{1}{2} \bar{q} C_\epsilon \begin{pmatrix} 0 & -I \\ I^\dagger & 0 \end{pmatrix} \bar{q}^{\bar{T}} \right\rangle =$$

$$= \left\langle -\frac{1}{2} \hat{q}^T C_\epsilon \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \hat{q} + \frac{1}{2} \bar{q} C_\epsilon \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix} \bar{q}^{\bar{T}} \right\rangle =$$

$$= \left\langle \frac{1}{2} \bar{q} C_\epsilon J_A \hat{q} + \frac{1}{2} \bar{q} C_\epsilon J_A \bar{q}^{\bar{T}} \right\rangle$$

this is like in Joachim (2.24): $\chi_L \equiv 2 \langle \bar{q} q \rangle$

Transformation

$\hat{q}$ transforms as q_R , i.e. $h \in SU(2)$

$$q_R \rightarrow h q_R$$

$$\hat{q} \rightarrow \hat{g} \hat{q}$$

$$\hat{h} = \begin{pmatrix} h & 0 \\ 0 & h \end{pmatrix}$$

Flavor symmetry: $\mathcal{L}_{kin} \supset \bar{\hat{q}} i \not{D}_\mu \hat{q}$

$$\Rightarrow \text{if } \hat{g} \in SU(2N_f) \Rightarrow$$

$$\hat{q} \rightarrow \hat{g} \hat{q}$$

$$\hat{q}^{\bar{T}} \rightarrow \hat{q}^{\bar{T}} \hat{g}^\dagger$$

$$\Rightarrow \mathcal{L}_{kin} = \text{inv}$$

$$\text{condensate: } \hat{q}^T J q \rightarrow \hat{q}^T \hat{g}^T J \hat{g} q = \text{inv if}$$

$$\hat{g}^T J \hat{g} = J \Rightarrow \hat{g} \in Sp(2N_f) \subset SU(2N_f)$$

For ^{unbroken} generators Q^a this means: $\hat{g} = e^{i\varepsilon^a Q^a}$, $Q^a \in su(2N_f)$

$$(1 + i\varepsilon^a Q^a)^T J (1 + i\varepsilon^a Q^a) = J$$

$$\Rightarrow i\varepsilon^a Q^{aT} J + i\varepsilon^a J Q^a = 0$$

$$\Rightarrow Q^{aT} J + J Q^a = 0$$

(in other convention
 $J' = -J$ but
 this definition
 does not change)

Back to condensates:

We have $\langle \bar{q} q \rangle = \frac{1}{2} \langle \hat{q}^T C \varepsilon J \Lambda \hat{q} + h.c. \rangle$

Bjerrum: $\langle \bar{q} q \rangle = \langle \hat{q}^T \varepsilon C J \hat{q} + h.c. \rangle$

Where is T & where
 is $1/2$?!

] Kulkarni, ^{N_f} "2 Dirac", pseudoreal fermions $\Rightarrow 2N_f = 4$ Weyl Dfs.

$$\mathcal{H}_{min} = (m_u + m_d) \mu^3$$

$$M_\pi^2 = \frac{\mu^3}{2f_\pi^2} (m_u + m_d)$$

$$\langle \bar{u} u \rangle = \frac{\partial \mathcal{H}}{\partial m_u} = \mu^3$$

$$\langle \bar{d} d \rangle = \frac{\partial \mathcal{H}}{\partial m_d} = \mu^3$$

total
 $\Rightarrow \langle \bar{q} q \rangle = 2\mu^3$

This is same as Bjerrum

$$-i q_L^T \gamma^0 \gamma^2 \gamma^0 \cdot \underbrace{\varepsilon \gamma^0}_{1} \not{D} \underbrace{\varepsilon \gamma^2 \gamma^0 \gamma^0}_{1} q_L^* = -i q_L^T \gamma^0 \gamma^2 \varepsilon \not{D} \varepsilon \gamma^2 q_L^* \quad \textcircled{=}$$

$$\not{D} = \gamma^\mu D_\mu = \gamma^\mu (\partial_\mu - i G_\mu) = \gamma^\mu (\partial_\mu - i G_\mu^a \phi^a)$$

$$\Rightarrow \varepsilon (\partial_\mu - i G_\mu^a \phi^a) \cdot \varepsilon = -\partial_\mu - i G_\mu^a \phi^a$$

$$\boxed{\varepsilon \vec{\partial} \varepsilon = \vec{\partial}^T}$$

$$\textcircled{=} -i q_L^T \gamma^0 \gamma^2 \cdot \gamma^\mu (-\partial_\mu - i G_\mu^a \phi^a) \gamma^2 q_L^* \quad \textcircled{=}$$

$$\gamma^2 \gamma^\mu \gamma^2 = \begin{cases} \mu=2 \Rightarrow \gamma^2 \gamma^2 \gamma^2 = -\gamma^2 = -\gamma^\mu \\ \mu \neq 2 \Rightarrow \gamma^2 \gamma^\mu \gamma^2 = \underbrace{-\gamma^2 \gamma^2}_{=-1} \gamma^\mu = \gamma^\mu \end{cases}$$

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

$$(\gamma^+ = \gamma^- \Rightarrow \gamma^{*+} = \gamma^{-T})$$

$$\text{i.e.} = (-1)^{\delta_{\mu 2}} \gamma^\mu = \gamma^{\mu*}$$

(γ_2 is the only
:neg: many)

$$\textcircled{=} -i q_L^T \gamma^0 \cdot \gamma^{\mu*} (-\partial_\mu - i G_\mu^a \phi^a) q_L^* =$$

$$= i q_L^T \gamma^0 \gamma^{\mu*} \partial_\mu q_L^* + i q_L^T \gamma^0 \gamma^{\mu*} i G_\mu^a \phi^a q_L^* =$$

$$= \underbrace{-i \partial_\mu q_L^T \gamma^0 \gamma^{\mu*} q_L^*}_{\textcircled{1}} + \underbrace{i q_L^T \gamma^0 \gamma^{\mu*} i G_\mu^a \phi^a q_L^*}_{\textcircled{2}} =$$

$$\textcircled{1} -i \partial_\mu q_L^T \gamma^0 \gamma^{\mu*} q_L^* = -i \partial_\mu q_L^\alpha \gamma^{0\alpha\beta} \gamma^{\mu* \beta\gamma} q_L^{*\gamma} = \left\{ \begin{array}{l} \text{terms anticommute} \Rightarrow \\ q_L^\alpha q_L^{*\gamma} = -q_L^{*\gamma} q_L^\alpha \end{array} \right\} =$$

$$= +i q_L^{*\gamma} \gamma^{0\alpha\beta} \gamma^{\mu* \beta\gamma} \partial_\mu q_L^\alpha =$$

$$= i q_L^{*\gamma} \gamma^{\mu* \beta\gamma} \gamma^{0\alpha\beta} \partial_\mu q_L^\alpha = i q_L^{*\gamma} (\gamma^{\mu*})^{\gamma\beta} \gamma^{0\beta\alpha} \partial_\mu q_L^\alpha$$

$$= i q_L^{*\gamma} \gamma^{\mu*} \gamma^0 \partial_\mu q_L = i q_L^{*\gamma} \gamma^0 \gamma^{\mu*} \gamma^0 \partial_\mu q_L = i q_L^{*\gamma} \gamma^0 \gamma^{\mu*} \gamma^0 \partial_\mu q_L = i \bar{q}_L \gamma^\mu \partial_\mu q_L \quad \checkmark$$

$$\gamma^{\mu*} = \gamma^0 \gamma^\mu \gamma^0$$

$$\gamma^{0*} = \gamma^0$$

$$= -\frac{1}{2} \left[-\bar{q}_{R\alpha} C_{\alpha\beta} \mathcal{M} \tilde{q}_{R\beta}^T + \bar{q}_{R\alpha} C_{\alpha\beta} \mathcal{M}^T \bar{q}_{R\beta}^T \right] =$$

$$= -\frac{1}{2} \left[-\bar{q}_{R\alpha} C_{\alpha\beta} \mathcal{M} \cdot \overline{(\varepsilon_{\beta\gamma} C_{\gamma\delta} \bar{q}_{L\delta}^T)}^T + \overline{(\varepsilon_{\alpha\gamma} C_{\gamma\delta} \bar{q}_{L\delta}^T)} C_{\alpha\beta} \mathcal{M}^T \bar{q}_{R\beta}^T \right] \textcircled{=}$$

$$\cdot \overline{(\varepsilon C \bar{q}_L^T)}^T = \overline{(\varepsilon C (q_L^+ \gamma^0)^T)}^T = \overline{(\varepsilon C \gamma^0 q_L^*)}^T = \left[(\varepsilon C \gamma^0 q_L^*)^+ \gamma^0 \right]^T =$$

$$= (q_L^T \gamma^0 C^+ \varepsilon^+ \gamma^0)^T = \gamma^0 \varepsilon^* C^* \gamma^0 q_L = \gamma^0 \varepsilon C^* \gamma^0 q_L$$

$$\cdot \bar{q}_R^T = (q_R^+ \gamma^0)^T = \gamma^0 q_R^*$$

What do we need the γ^0 in does for?!!

$$\textcircled{=} -\frac{1}{2} \left[-\bar{q}_R C_{\alpha\beta} \mathcal{M} \cdot \underbrace{\gamma^0 \varepsilon C^* \gamma^0 q_L}_{-1} + \underbrace{q_L^T \gamma^0 C^+ \gamma^0}_{-1} \underbrace{\varepsilon \mathcal{M}^T \gamma^0 q_R^*}_{+1} \right]$$

$$= -\frac{1}{2} \left[\bar{q}_R C \mathcal{M} \gamma^0 C^* \gamma^0 q_L + q_L^T \gamma^0 C^+ \gamma^0 C \mathcal{M}^T \gamma^0 q_R^* \right] \textcircled{=}$$

$$\cdot C \gamma^0 C^* \gamma^0 = \underbrace{i \gamma^2 \gamma^0 \gamma^0}_{=1} \underbrace{(-i \gamma^2 \gamma^0)^*}_{=1} \gamma^0 = \gamma^2 \gamma^2 \gamma^0 \gamma^0 = \gamma^2 \gamma^2 = \overbrace{\gamma^2 \gamma^2}^{-1} \overbrace{\gamma^2 \gamma^2}^{-1} = 1$$

$$\gamma^{\mu*} = \gamma^2 \gamma^\mu \gamma^2$$

$$\cdot C^+ \gamma^0 C \gamma^0 = -i (\gamma^2 \gamma^0)^+ \gamma^0 (i \gamma^2 \gamma^0) \gamma^0 = \gamma^0 \gamma^2 \gamma^0 \gamma^2 \gamma^0 \gamma^0 = \underbrace{\gamma^0 \gamma^0}_{=1} \underbrace{\gamma^2 \gamma^2}_{=1} \underbrace{\gamma^0 \gamma^0}_{=1} = -1$$

$$\textcircled{=} -\frac{1}{2} \left[\bar{q}_R \mathcal{M} q_L - q_L^T \gamma^0 \mathcal{M}^T q_R^* \right] \Rightarrow -\frac{1}{2} \left[\bar{q}_R \mathcal{M} q_L + \bar{q}_R \mathcal{M} q_L \right] = \bar{q}_R \mathcal{M} q_L \checkmark$$

$$\left\{ \begin{aligned} q_L^T \gamma^0 \mathcal{M}^T q_R^* &= q_L^{\mu} \gamma^0_{\mu\nu} \mathcal{M}^{\nu\alpha} q_R^{*\alpha} = -q_R^{\alpha} \gamma^0_{\alpha\beta} \mathcal{M}^{\beta\mu} q_L^{\mu} = -q_R^{\alpha} \gamma^0_{\alpha\beta} \mathcal{M}^{\beta\mu} q_L^{\mu} = -q_R^{\alpha} \mathcal{M} q_L^{\mu} = \\ &= -\bar{q}_R \mathcal{M} q_L \end{aligned} \right\}$$

$$\text{Bijlens: } m_\pi^2 = B_0(m_u + m_d) \Rightarrow B_0 = \frac{\mu^3}{2f_\pi^2} \Rightarrow \mu^3 = 2f_\pi^2 \cdot B_0$$

$$\Rightarrow \langle \bar{q}q \rangle = 2 \cdot 2 f_\pi^2 B_0 = 4 f_\pi^2 B_0 = \underline{2N_f}$$

$$\text{Bijlens has } \langle \bar{q}q \rangle = \frac{2f_\pi^2 B_0}{N_f}$$

$$\langle \bar{\psi}_{L,i} \psi_{R,j} \rangle = \frac{\delta_{ij}}{2N_f} \langle \bar{q}q \rangle$$

$$\Rightarrow \langle \bar{q}q \rangle = \sum_i \langle \bar{\psi}_{L,i} \psi_{R,i} \rangle + \langle \bar{\psi}_{R,i} \psi_{L,i} \rangle = \sum_i \left(\frac{\langle \bar{q}q \rangle}{2} \delta_{ii} + \delta_{ii} \frac{\langle \bar{q}q \rangle}{2} \right) = \underline{N_f} \langle \bar{q}q \rangle$$

QCD

$$\langle \bar{q}q \rangle = \langle \bar{q}_R q_L \rangle + \text{h.c.}$$

pseudoscal, $SU(N_c=2)$

$$\text{we: } \langle \bar{q}q \rangle = \frac{1}{2} \langle \hat{q}^T C \varepsilon J_A \hat{q} + \text{h.c.} \rangle$$

$$\text{Bijlens: } \langle \bar{q}q \rangle = \langle \hat{q}^T C \varepsilon J_A \hat{q} \rangle + \text{h.c.}$$

$$\langle \bar{q}_{L,i} q_{R,i} \rangle = \frac{1}{2} \langle \bar{q}q \rangle \delta_{ij} \quad \text{denote this by } \delta \text{ to be convenient.}$$

$$\langle \bar{q}q \rangle = \sum_{i=1, N_f} \langle \bar{q}_{R,i} q_{L,i} + \bar{q}_{L,i} q_{R,i} \rangle = -N_f B_0 F^2$$

↑
Dirac fermions.
⇒ 2N_F ways.

$$\langle \hat{q}_i^T C \varepsilon q_j \rangle = \frac{1}{2} \langle \bar{q}_L q_R \rangle J_{A,ij}$$

What do I calculate? I take $\frac{\partial \mathcal{H}_{eff}}{\partial m_u} \Big|_{\min}$, $\frac{\partial \mathcal{H}_{eff}}{\partial m_d}$

⇒ I really just calculate $\langle \bar{q}q \rangle$

$$\begin{aligned} \langle \bar{q}q \rangle &= \sum_{i=1}^{N_f} \langle \bar{q}_{L,i} q_{R,i} \rangle + \langle \bar{q}_{R,i} q_{L,i} \rangle = \\ &= 2 \sum_{i=1}^{N_f} \langle \bar{q}_{L,i} q_{R,i} \rangle = 2 \sum_{i=1}^{N_f} \frac{1}{2} \delta \delta_{ii} = \delta \sum_{i=1}^{N_f} \delta_{ii} = \delta N_f \end{aligned}$$

$$\begin{aligned} \langle \bar{q}q \rangle &= \sum_{i=1}^{N_f} \langle \bar{q}_{L,i} q_{R,i} \rangle + \langle \bar{q}_{R,i} q_{L,i} \rangle = \\ &= 2 \sum_{i=1}^{N_f} \langle \bar{q}_{L,i} q_{R,i} \rangle = 2 \cdot \sum_{i=1}^{N_f} \left(-\frac{1}{2} \hat{q}^T C \varepsilon J^+ \hat{q} \right) = \end{aligned}$$

$\Rightarrow \delta = \langle \bar{q}_i q_i \rangle$, no sum!
for single flavour

$\hat{q} =$

$\hat{q}_i^T C q_j =$
... second term is

$$\text{Dirac} = \bar{\psi}(i \not{D} - m)\psi$$

$$\hat{q}_i^T \varepsilon C \hat{q}_j = (q_{R_i}^T, \tilde{q}_{R_i}^T) \varepsilon C \begin{pmatrix} q_{R_j} \\ \tilde{q}_{R_j} \end{pmatrix} =$$

$$= \begin{pmatrix} q_{R_i}^T, & \bar{q}_{L_i} C^T \varepsilon^T \end{pmatrix} \varepsilon C \begin{pmatrix} q_{R_j} \\ \varepsilon C \bar{q}_{L_j}^T \end{pmatrix} =$$

$$q_R^T \gamma^2 \gamma q_R$$

$$= \begin{pmatrix} q_{R_i}^T, & \bar{q}_{L_i} C^T \varepsilon^T \end{pmatrix} \begin{pmatrix} \varepsilon C q_{R_j} \\ \varepsilon C \varepsilon C \bar{q}_{L_j}^T \end{pmatrix} = q_{R_j}^T \varepsilon C q_{R_j} + \bar{q}_{L_i} C^T \varepsilon^T \varepsilon C \varepsilon C \bar{q}_{L_j}^T$$

$\parallel$
 $(q_{L_j}^T)^T =$
 $= \gamma^0 q_{L_j}^\alpha$

$$= q_{R_j}^T \varepsilon C q_{R_j} +$$

$$\square \otimes \square = \begin{bmatrix} \square \\ \square \end{bmatrix} \oplus \begin{bmatrix} \square & \square \end{bmatrix}$$

$$\psi^T J \psi \rightarrow \psi^T U^T J U \psi$$

d_f - color i - flavor

μ - Lorentz

1) pseudo-real case:

$$\psi_{i\alpha}^T \delta_2^{\mu\nu} T_A^{ij} \psi_j^\beta \rightarrow \psi_{\textcircled{i}}^T \textcircled{\alpha} \textcircled{\mu} T_A^{ij} \psi_{\textcircled{j}}^{\textcircled{\beta} \textcircled{\nu}} =$$

↑ antisymmetric
 $\Rightarrow$ we need to take
the antisymmetric combination
of ij !

antisymmetric
representation

2) real case: $\psi_i^T \delta_2^{ij} J_S^{ij} \psi_j \propto \psi_i^T \delta_2^{ij} J_S^{ij} \psi_j$

$$\langle \bar{\psi}_i \psi_j \rangle \sim \delta_{ij}$$

singlet of
the flavor
vector symmetry
(if δ_2 is
in image of
the flavor
symmetry)

$$\langle \bar{\psi}_i \psi_j \rangle$$

sharp

lower to

singlet +

component

in reps. representation.

(unbroken symmetry!)



Singlet w.r.t

Brauer EFT

Chapter 3

Consider vacuum: $\langle \phi \rangle \equiv \phi_0$

broken generators: $T_a \phi_0 \neq 0$

unbroken: $T_\alpha \phi_0 = 0$

Field parametrization: $\phi(x) = U(x) [\phi_0 + H(x)]$

↑
integrate this out

can choose $U(x) = e^{i\theta^a(x)T_a}$

$\phi^\dagger \phi$ do not have $U(x)$. But $\partial_\mu \phi = U \{ \partial_\mu H + (U^{-1} \partial_\mu U) (\phi_0 + H) \}$

$U^{-1} \partial_\mu U \in$ algebra of the symmetry. $\Rightarrow$ can use group th. for Lagrangian

Coxeter construction

We have SSB: $G \rightarrow H$. How do GBs transform under G ?

We consider: $\phi = U \phi_0 \xrightarrow{g \in G} g\phi = gU\phi_0$
(H already int. out)

Can gU be written as $e^{i\theta'^a T_a}$? Not always!

(If we could do this then we would have transformation law

$$U = e^{i\theta^a T_a} \rightarrow gU = e^{i\theta'^a T_a} \Rightarrow \text{i.e. } \theta'^a(\theta^a)$$

But: $h\phi_0 = \phi_0, h \in H$

At least for infinitesimal transformation we can decompose:

$$gU = U' h = \underbrace{e^{i\theta'^a T_a}}_{U'(\theta')} \underbrace{e^{i\varepsilon^\alpha T_\alpha}}_{h} \leftarrow \text{unbroken}$$

But why can we decompose like this?

$\Rightarrow$

$$U'(\theta) = g U(\theta) h^{-1}(g, \theta)$$

- non-linear realization of global symmetry G .

It defines how G acts on GBs.

g - does not depend on x - global symmetry; but $h = h(x)$

Our building block is Maurer - Cartan form

$$\omega_\mu = -i U^{-1} \partial_\mu U$$

$$\begin{aligned} \omega_\mu &\xrightarrow{g} -i h U^{-1} g^{-1} \partial_\mu (g U h^{-1}) = -i h U^{-1} g^{-1} g (\partial_\mu U h^{-1} + U \partial_\mu h^{-1}) = \\ &= -i h U^{-1} \partial_\mu U h^{-1} - i h U^{-1} U \partial_\mu h^{-1} = h \omega_\mu h^{-1} - \underbrace{i h \partial_\mu h^{-1}}_{\in \mathfrak{h}} \end{aligned}$$

We can decompose ω_μ :

$$\omega_\mu = \underbrace{\omega_\mu^{\perp a} T_a}_{\substack{\omega_\mu^\perp \\ \text{broken part}}} + \underbrace{\omega_\mu^{\parallel \alpha} T_\alpha}_{\substack{\omega_\mu^\parallel \\ \text{unbroken part, } \in \mathfrak{h}}}$$

$$\Rightarrow \begin{cases} \omega_\mu^\perp \xrightarrow{g} h \omega_\mu^\perp h^{-1} \\ \omega_\mu^\parallel \xrightarrow{g} h \omega_\mu^\parallel h^{-1} - i h \partial_\mu h^{-1} \end{cases} \quad \text{— matter fields}$$

gauge transformations from $H \Rightarrow$ "gauge field"

$$\mathcal{L}_{\text{eff}} = \frac{1}{2} g_{ab} \omega^{\perp a \mu} \omega_\mu^{\perp b}$$

↑
symmetric invariant tensor of H

Symmetric coset spaces

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{k}$$

↑
spanned by broken generators

• $[\mathfrak{h}, \mathfrak{h}] \in \mathfrak{h}$

• $[\mathfrak{h}, \mathfrak{k}] \in \mathfrak{k}$ (broken generators span a representation of H)

• $[\mathfrak{k}, \mathfrak{k}] \in \mathfrak{h}$

In this case Lie algebra is invariant under automorphisms
 $T_i \rightarrow R(T_i) : \begin{cases} T_a \rightarrow T_a \\ T_a \rightarrow -T_a \end{cases} \quad (\alpha - \text{unbroken}, a - \text{broken})$

We had: $U(\theta) = e^{i\theta^a T_a} \quad \& \quad U' = g U h^{-1}$

$$U'^{-1} = U'^{\dagger} = e^{-i\theta^a T_a^{\dagger}} = e^{-i\theta^a T_a} = R(U') = R(g U h^{-1}) =$$

$$\Rightarrow = R(g) R(U) R(h^{-1}) = R(g) U^{-1} h^{-1}$$

$$\Rightarrow U'^{-1} = R(g) U^{-1} h^{-1} \Rightarrow U' = h U R(g)^{-1}$$

Take $\Sigma(\theta) \equiv U(\theta)^2 \rightarrow$

We now use both: $U' = g U h^{-1} \quad \& \quad U' = h U R(g)^{-1}$

$$\Sigma(\theta) = U^2 \rightarrow \underbrace{g U h^{-1}} \cdot \underbrace{h U R(g)^{-1}} = g U^2 R^{-1}(g) = g \underbrace{\Sigma R^{-1}(g)}_{\equiv \Sigma'}$$

No dependence on coordinates; linear transformation

Chapter 5

ChPT for QCD

$$G = SU(N_f)_L \times SU(N_f)_R \xrightarrow{\text{SSB}}$$

$$H = SU(N_f)_V$$

$$g = (g_L, g_R) \in G$$

$$g_L = g_R \Rightarrow h = (h, h)$$

Broken generators: axial generators that transform ψ_L & ψ_R in the opposite way $\Rightarrow$ we can write our matrix U as:

$$U \equiv (u, u^{-1}), \quad u \in SU(N_f)$$

We had $U' = g U h^{-1} \Rightarrow$

$$U' = (u', u'^{-1}) = (g_L, g_R) (u, u^{-1}) (h^{-1}, h^{-1})$$

$\Rightarrow$ we have separately:

$$\begin{cases} U' = g_L u h^{-1} \\ u'^{-1} = g_R u^{-1} h^{-1} \Rightarrow U' = h u g_R^{-1} \end{cases}$$

$$\Rightarrow U'^2 = U' U' = g_L u h^{-1} \cdot h u g_R^{-1} = g_L U^2 g_R$$

We found a variable which transforms linearly under G :

$$\Sigma \equiv U^2 \Rightarrow \Sigma' = g_L \Sigma g_R$$

] Maurer-Cartan form: $i\omega_\mu = U^{-1} \partial_\mu U$

$$\begin{aligned} i\omega_\mu = U^{-1} \partial_\mu U &= (u^{-1}, u) \partial_\mu (u, u^{-1}) = (\bar{u}^T, u) \left((\partial_\mu u, u^{-1}) + (u, \partial_\mu u^{-1}) \right) = \\ &= (u^{-1} \partial_\mu u, \mathbb{1}) + (\mathbb{1}, u \partial_\mu u^{-1}) \end{aligned}$$

We need to consider the broken part: $\omega_\mu^\perp = c_{\mu a}^\perp T^a$
 $\uparrow$ broken generators

Using $R(T_i)$ we can project out $\omega^\perp$:

$$i\omega_\mu^\perp = \frac{1}{2} (i\omega_\mu - R(i\omega_\mu)) \quad \textcircled{=}$$

$$R(U) = R(u, u^{-1}) = R(u^{-1}, u) \quad (R(g_L, g_R) = R(g_R, g_L))$$

$$\Rightarrow R(i\omega_\mu) = (u, u^{-1}) \partial_\mu (u^{-1}, u) = (u \partial_\mu u^{-1}, \mathbb{1}) + (\mathbb{1}, u^{-1} \partial_\mu u)$$

$$\textcircled{=} \frac{1}{2} \left((u^{-1} \partial_\mu u, \mathbb{1}) + (\mathbb{1}, u \partial_\mu u^{-1}) - (u \partial_\mu u^{-1}, \mathbb{1}) - (\mathbb{1}, u^{-1} \partial_\mu u) \right)$$

$$= \frac{1}{2} \left(\underbrace{(u^{-1} \partial_\mu u - u \partial_\mu u^{-1})}_{X_\mu}, 1 \right) + \left(1, \underbrace{u \partial_\mu u^{-1} - u^{-1} \partial_\mu u}_{-X_\mu} \right)$$

$$X_\mu = u^{-1} \partial_\mu u - u \partial_\mu u^{-1} = \left\{ 0 = \partial_\mu (u u^{-1}) = u \partial_\mu u^{-1} + \partial_\mu u u^{-1} \right\} =$$

where is 2

?

$$= u^{-1} \partial_\mu u + \partial_\mu u \cdot u^{-1} = u^{-1} \partial_\mu u^2 u^{-1} = u^{-1} \partial_\mu \Sigma u^{-1}$$

or alternatively: $= u \partial_\mu \Sigma^{-1} u$

$$\Rightarrow \text{do } \text{tr}(\omega_\mu^\dagger \omega_\mu^\dagger) = \text{tr}(u^{-1} \partial_\mu \Sigma u^{-1} \cdot u \partial_\mu \Sigma^{-1} u) = \text{tr}(\partial_\mu \Sigma \partial^\mu \Sigma^{-1})$$

Chapter 7 QCD-like theories

In QCD-like theories the kinetic term is invariant under

$G = SU(2N_f)$ while condensate & mass term only under:

$$h \in Sp(2N_f) : h^T \begin{pmatrix} 0 & 1 \\ \pm 1 & 0 \end{pmatrix} h = \begin{pmatrix} 0 & 1 \\ \pm 1 & 0 \end{pmatrix} \Rightarrow Sp(2N_f) \text{ or } SO(2N_f)$$

For $Sp \Rightarrow \Sigma_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

for $SO \Rightarrow \Sigma_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Broken generators:

$$\Pi^T \Sigma_0 - \Sigma_0 \Pi = 0 \Rightarrow \Pi^* = \Sigma_0^{-1} \Pi \Sigma_0$$

Unbroken generators:

$$\Pi^T \Sigma_0 + \Sigma_0 \Pi = 0 \Rightarrow \Pi^* = -\Sigma_0^{-1} \Pi \Sigma_0$$

Chapter 8 EFT for QCD-like theories

We had $\Sigma_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ $1 \equiv 1_{N_f \times N_f}$

Generally,

$$U \rightarrow g U^{-1} h$$

made of broken generators $U = e^{i \theta^a T^a}$

$$\bullet h \in H \Rightarrow h = e^{i a^d T^d} \Rightarrow \Sigma_0^{-1} h \Sigma_0 = \Sigma_0^{-1} e^{i a^d T^d} \Sigma_0 =$$

$$= \Sigma_0^{-1} (1 + i a^d T^d) \Sigma_0 = 1 + i a^d \Sigma_0^{-1} T^d \Sigma_0 = 1 - i a^d T^{d*} =$$

$$= e^{-i a^d T^{d*}} = (e^{i a^d T^d})^* = h^*, \text{ i.e. } h^* = \Sigma_0^{-1} h \Sigma_0$$

• for U we have

$$\Sigma_0^{-1} U^{-1} \Sigma_0 = \Sigma_0^{-1} e^{-i \theta^a T^a} \Sigma_0 = \Sigma_0^{-1} (1 - i \theta^a T^a) \Sigma_0 =$$

$$= 1 - i \theta^a T^{a*} = e^{-i \theta^a T^{a*}} = U^* \Rightarrow U^* = \Sigma_0^{-1} U^{-1} \Sigma_0$$

$$\Rightarrow U = \Sigma_0^{-1} U^{-1*} \Sigma_0 = \varepsilon \Sigma_0 U^{-1*} \varepsilon \Sigma_0^{-1} = \Sigma_0 U^{-1*} \Sigma_0^{-1} \Rightarrow U^{-1*} = \Sigma_0^{-1} U \Sigma_0$$

$$\begin{pmatrix} 0 & 1 \\ \varepsilon & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ \varepsilon & 0 \end{pmatrix} = \begin{pmatrix} \varepsilon & 0 \\ 0 & \varepsilon \end{pmatrix} = \varepsilon \mathbb{1} \Rightarrow \left[\varepsilon \begin{pmatrix} 0 & 1 \\ \varepsilon & 0 \end{pmatrix} \right] \cdot \Sigma = \mathbb{1}$$

$$\Rightarrow \Sigma^{-1} = \varepsilon \Sigma$$

Consider now:

$$\Sigma = U^2 \Sigma_0$$

$$\bullet \Sigma^+ \Sigma = \Sigma_0^+ (U^2)^+ U^2 \Sigma_0 = \Sigma_0^+ \overbrace{U^+ U^+ U U}^{\mathbb{1}} \Sigma_0 = \Sigma_0^+ \Sigma_0 = \mathbb{1} \Rightarrow \Sigma \text{ is unitary}$$

$$U \in \text{SU}(2N_f)$$

$$\Sigma_0^+ \Sigma = \begin{pmatrix} 0 & \varepsilon \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ \varepsilon & 0 \end{pmatrix} = \begin{pmatrix} \varepsilon^2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\bullet \Sigma^T = (U^2 \Sigma_0)^T = \Sigma_0^T U^T U^T = \varepsilon \Sigma_0 (U^{-1})^* (U^{-1})^+ = \varepsilon \Sigma_0 \underbrace{(\Sigma_0^{-1} U \Sigma_0)}_{\mathbb{1}} \underbrace{(\Sigma_0^{-1} U \Sigma_0)}_{\mathbb{1}} \quad \textcircled{=}$$

$$U^+ = U^{-1} \Rightarrow U^T = (U^{-1})^*$$

$$\textcircled{=} \varepsilon U^2 \Sigma_0 \Rightarrow \text{for } S_p - \text{antisymmetric}$$

$$\text{for } SO - \text{symmetric}$$

transformation

$$\Sigma = U^2 \Sigma_0 = U (\Sigma_0 U^{-1*} \Sigma_0^{-1}) \Sigma_0 = U \Sigma_0 U^{-1*} \rightarrow (g U h^{-1}) \Sigma_0 h^* U^{-1*} g^{-1*}$$

$$\stackrel{1}{=} (g U h^{-1}) \Sigma_0 \underbrace{(\Sigma_0^{-1} h \Sigma_0^{-1} U \Sigma_0 g^T)}_{1} = g U^2 \Sigma_0 g^T = g \Sigma g^T$$

$$g^+ = g^{-1} \Rightarrow g^T = (g^{-1})^+$$

$$\Rightarrow \det \propto \text{tr} (2_\mu \Sigma \Sigma^\mu \Sigma^+) \rightarrow \text{tr} (g 2_\mu \Sigma g^T \underbrace{g^{T+} 2_\mu \Sigma^+ g^+}_{(g^+ g)^T = 1}) = \text{inv.}$$

$$\Sigma = U^2 \Sigma_0 = U U \Sigma_0 = U (1 + i a^a T^a) \Sigma_0 = U (\Sigma_0 + i a^a T^a \Sigma_0) \quad \textcircled{E}$$

(Broken generators: $\Pi^T \Sigma_0 - \Sigma_0 \Pi = 0 \Rightarrow T_a^T \Sigma_0 = \Sigma_0 T^a \Rightarrow$

$$\Rightarrow \Sigma_0^{-1} T_a^T \Sigma_0 = T^a$$

$$\Rightarrow \Sigma_0^{-1} T_a^T = T^a \Sigma_0^{-1}$$

$$\Sigma_0^{-1} = \varepsilon \Sigma_0$$

$$\Rightarrow \cancel{\varepsilon} \Sigma_0 T_a^T = T_a \cancel{\varepsilon} \Sigma_0 \Rightarrow T_a \Sigma_0 = \Sigma_0 T_a^T$$

$$\textcircled{E} \quad U (\Sigma_0 + i a^a \Sigma_0 T_a^T) = U \Sigma_0 (1 + i a^a T_a^T) = U \Sigma_0 U^T$$

Pompeu: $\chi_c \equiv 2 \langle \bar{q} q \rangle$

$f_{\text{Pompeu}} = F_{\text{Bjorken}}$

$$m_\pi^2 = \frac{m \chi_c}{4f^2} = \frac{m}{4f^2} 2 \langle \bar{q} q \rangle = \frac{m \langle \bar{q} q \rangle}{2f^2}$$

Bjorken:

$$\langle \bar{q} q \rangle = 2 N_f B_0 F^2 \quad (5.1)$$

$$\Rightarrow \langle \bar{q} q \rangle = 2 B_0 F^2 \cdot 2$$

$$M^2 = 2 B_0 m$$

$$\Rightarrow M^2 = 2m \cdot \frac{\langle \bar{q} q \rangle}{2F^2 \cdot 2} = \frac{m \langle \bar{q} q \rangle}{2F^2}$$

We need additional 2:
to make results the same!

General case:

$$d = \bar{q}_L i \gamma^\mu D_\mu q_L + \bar{q}_R i \gamma^\mu D_\mu q_R - \bar{q}_R M_{ij} q_L - \bar{q}_L M_{ij}^+ q_R, \quad q_L \equiv \begin{pmatrix} q_L \\ 0 \end{pmatrix}, \quad q_R \equiv \begin{pmatrix} 0 \\ q_R \end{pmatrix}$$

We have here $2N_f$ Weyl d.o.f.

$$D_\mu q = \partial_\mu q - i G_\mu q$$

$$G_\mu = G_\mu^a T^a$$

$$T^{a\dagger} = T^a$$

- gauge group generators

$$q \rightarrow e^{i d^a T^a} q$$

Note: $\text{Kogut} \supset SU(N_c)$ group & then for $N_c=2 \Rightarrow$ fundamental is pseudoreal, & for $\forall N_c$ adjoint is real. We will just as an example have fundamental representation of $Sp(N_c) \Rightarrow$ pseudoreal & vector rep of $SO(N_c) \Rightarrow$ real. (see Peskin, table 3).

Note if we can write:

$$U^* = S U S^{-1} \quad U = e^{i d^a T^a}$$

$\uparrow \quad \uparrow$ matrices in color space

$$S^\dagger S = \mathbb{1} \quad \text{unitary}$$

$$\Rightarrow 1 - i d^a T^{a*} = S (1 + i d^a T^a) S^{-1}$$

$$\Rightarrow -i d^a T^{a*} = i d^a S T^a S^{-1} \Rightarrow -T^{a*} = S T^a S^{-1}$$

$$(\text{if } T^{a\dagger} = T^a \Rightarrow T^{aT} = T^{a*} \Rightarrow -T^{aT} = S T^a S^{-1}) \quad (\text{Peskin 2.4})$$

If S -symm $\Rightarrow$ rep is real

S -antisymm $\Rightarrow$ rep is pseudoreal

Invers has same properties

$$S^{-1} S = \mathbb{1}$$

$$S^T (S^{-1})^T = \mathbb{1} \Rightarrow S (S^{-1})^T = \mathbb{1} \Rightarrow (S^{-1})^T = S^{-1}$$

We define new object: $C = i \gamma^2 \gamma^0$

$$\tilde{q}_{R\alpha,i} = \epsilon_{\alpha\beta} C_{\beta\gamma}^{-T} q_{L\beta,i}$$

$$\epsilon \equiv S^{-1}$$

ϵ have same symmetry/antisym. properties as S

here α, β - group indices (color)

$$P_L \tilde{q}_R = 0, \quad P_R \tilde{q}_R = \tilde{q}_R \Rightarrow \text{right-handed}$$

How $\tilde{q}_R$ does transform?

(omit flavour index)

$$\begin{aligned}\tilde{q}_{Rd} &= \epsilon_{\alpha\beta} C \bar{q}_{L\beta}^T = \epsilon_{\alpha\beta} C (q_{L,\beta}^\dagger \gamma^0)^T = \\ &= \epsilon_{\alpha\beta} C \gamma^{0T} q_{L,\beta}^* \xrightarrow{\text{gauge}} \epsilon_{\alpha\beta} C \gamma^{0T} \left(e^{i\alpha^a T_a^*} \right)_{\beta\gamma} q_{L,\gamma}^* = \\ &= \epsilon_{\alpha\beta} C \gamma^{0T} \left(e^{-i\alpha^a T_a^*} \right)_{\beta\gamma} q_{L,\gamma}^* \quad \textcircled{=}\end{aligned}$$

expanding: $\epsilon_{\alpha\beta} (\delta_{\beta\gamma} - i\alpha^a T_{a,\beta\gamma}^*) q_{L,\gamma}^* \quad \textcircled{=}$

key moment: we use that fact that we can connect T^* with T !

We know: $-T_a^* = S T_a S^{-1} \Rightarrow$

$$\textcircled{=} \epsilon (1 + i\alpha^a \cdot S T_a S^{-1}) q_L^* \stackrel{\epsilon S = 1}{=} (S^{-1} + i\alpha^a T_a S^{-1}) q_L^* = (1 + i\alpha^a T_a) (\epsilon q_L^*) = e^{i\alpha^a T_a} \epsilon q_L^*$$

$$\textcircled{=} C \gamma^{0T} \cdot e^{i\alpha^a T_a} \epsilon q_L^* = e^{i\alpha^a T_a} C \gamma^{0T} \epsilon q_L^* = e^{i\alpha^a T_a} \underbrace{\epsilon C \cdot \bar{q}_L^T}_{\equiv \tilde{q}_R} \quad (q^\dagger \gamma^0)^T = \gamma^{0T}$$

$\Rightarrow \tilde{q}_R$ transforms same as q !

$\Rightarrow$ we can build a new object

$$\hat{q} = \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix}$$

N_f R-handed Dirac spinors (i.e. N_f Weyl d.o.fs)

$$\hat{q} \rightarrow e^{i\alpha^a T_a} \hat{q}$$

corresponds to

N_f L-handed Dirac spinors (i.e. N_f Weyl d.o.fs)

kinetic term

Let us denote

$\varepsilon^T = \eta \varepsilon$, $\eta = \begin{cases} 1, & \text{real} \\ -1, & \text{pseudoreal} \end{cases}$ terms

$$\bar{\hat{q}} i \gamma^\mu D_\mu \hat{q} = (\bar{q}_R \tilde{\bar{q}}_R) \cdot i \gamma^\mu D_\mu \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix} = (\bar{q}_R \tilde{\bar{q}}_R) \begin{pmatrix} i \gamma^\mu D_\mu q_R \\ i \gamma^\mu D_\mu \tilde{q}_R \end{pmatrix} =$$

$$= i \bar{q}_R \not{D} q_R + i \tilde{\bar{q}}_R \not{D} \tilde{q}_R$$

$$\tilde{q}_R = \varepsilon \cdot C \bar{q}_L^T \quad (C = i \gamma^2 \gamma^0)$$

$$\Rightarrow \tilde{\bar{q}}_R = (\tilde{q}_R)^+ \gamma^0 = (\varepsilon \cdot C \bar{q}_L^T)^+ \gamma^0 =$$

$$= (\bar{q}_L)^{T+} C^+ \varepsilon^+ \gamma^0 = (q_L^+ \gamma_0)^T C^+ \varepsilon^+ \gamma^0 =$$

$$= (\gamma_0^T q_L^{+T})^+ C^+ \varepsilon^+ \gamma^0 = \underbrace{q_L^T \gamma_0}_{\gamma_0} C^+ \varepsilon^+ \gamma^0 \underbrace{\varepsilon^{-1}}_{\rightarrow \text{unitary!}}$$

$$\Rightarrow i \tilde{\bar{q}}_R \not{D} \tilde{q}_R = i q_L^T \gamma_0 C^+ \varepsilon^+ \gamma^0 \not{D} \varepsilon C \cdot \bar{q}_L^T \quad \textcircled{=}$$

(sum over
flavour is suppressed)

$$C^+ = (i \gamma^2 \gamma^0)^+ = -i \gamma^{0+} \gamma^{2+} = -i \gamma^0 \cdot \gamma^0 \gamma^2 \gamma^0 = -i \gamma^2 \gamma^0 = -C$$

$$(\gamma^\mu)^+ = \gamma^0 \gamma^\mu \gamma^0$$

$$\textcircled{=} i q_L^T \gamma_0 \underline{(-i \gamma^2 \gamma^0)} \varepsilon^+ \gamma^0 \not{D} \varepsilon \underline{(i \gamma^2 \gamma^0)} \cdot (q_L^+ \gamma_0)^T =$$

$$= i q_L^T \gamma^0 \gamma^2 \gamma^0 \cdot \varepsilon^+ \gamma^0 \not{D} \varepsilon \gamma^2 \gamma^0 \gamma^0 q_L^{*T} = i q_L^T \gamma^0 \gamma^2 \varepsilon^+ \not{D} \varepsilon \gamma^2 q_L^{*T} \quad \textcircled{=}$$

Use: $\not{D} = \gamma^\mu D_\mu = \gamma^\mu (\partial_\mu - i G_\mu) = \gamma^\mu (\partial_\mu - i G_\mu^a T^a)$

We have $\begin{matrix} T^{a*} \\ -T^{aT} = \varepsilon^{-1} T^a \varepsilon \end{matrix}$

$$\Rightarrow \varepsilon^{-1} (\partial_\mu - i G_\mu^a T^a) \cdot \varepsilon = \partial_\mu + i G_\mu^a T^{a*}$$

$$\textcircled{=} i q_L^T \gamma^0 \gamma^2 \cdot \gamma^\mu (\partial_\mu + i G_\mu^a T^{a*}) \gamma^2 q_L^{*T} \quad \textcircled{=}$$

$$\gamma^2 \gamma^\mu \gamma^2 = \begin{cases} \mu=2 \Rightarrow \gamma^2 \gamma^2 \gamma^2 = -\gamma^2 = -\gamma^\mu \\ \mu \neq 2 \Rightarrow \gamma^2 \gamma^\mu \gamma^2 = -\gamma^2 \gamma^2 \gamma^\mu = \gamma^\mu \end{cases} \quad \text{i.e.} = (-1)^{\delta_{\mu 2}} \gamma^\mu = \gamma^{\mu*}$$

$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ $(\gamma_2 \text{ is the only negative})$

$$\textcircled{=} \quad i \bar{q}_L^T \gamma^0 \cdot \gamma^{\mu*} (\partial_\mu + i G_\mu^a T_a^*) q_L^* =$$

$$= i \bar{q}_L^T \gamma^0 \gamma^{\mu*} \partial_\mu q_L^* + i \bar{q}_L^T \gamma^0 \gamma^{\mu*} i G_\mu^a T_a^* q_L^* = \text{int by parts}$$

$$= \underbrace{-i \partial_\mu \bar{q}_L^T \gamma^0 \gamma^{\mu*} q_L^*}_{\textcircled{1}} + \underbrace{i \bar{q}_L^T \gamma^0 \gamma^{\mu*} i G_\mu^a T_a^* q_L^*}_{\textcircled{2}} =$$

$$\textcircled{1} \quad -i \partial_\mu \bar{q}_L^T \gamma^0 \gamma^{\mu*} q_L^* = -i \partial_\mu q_L^\alpha \gamma^{0\alpha\beta} \gamma^{\mu*\beta\gamma} q_L^{*\gamma} = \left\{ \begin{array}{l} \text{terms anticommute} \Rightarrow \\ q_L^\alpha q_L^{*\gamma} = -q_L^{*\gamma} q_L^\alpha \end{array} \right\} =$$

$$= +i q_L^{*\gamma} \gamma^{0\alpha\beta} \gamma^{\mu*\beta\gamma} \partial_\mu q_L^\alpha =$$

$$= i q_L^{*\gamma} \gamma^{\mu*\beta\gamma} \gamma^{0\alpha\beta} \partial_\mu q_L^\alpha = i q_L^{*\gamma} (\gamma^{\mu*})^\gamma_\beta (\gamma^0)^{\beta\alpha} \partial_\mu q_L^\alpha =$$

$$= i \bar{q}_L^+ \gamma^{\mu+} \gamma^0 \partial_\mu q_L = i \bar{q}_L^+ \gamma^0 \gamma^{\mu+} \gamma^0 \partial_\mu q_L = i \bar{q}_L^+ \gamma^{\mu+} \partial_\mu q_L = i \bar{q}_L \gamma^\mu \partial_\mu q_L \quad \checkmark$$

$$\gamma^{\mu+} = \gamma^0 \gamma^\mu \gamma^0$$

$$\gamma^{0T} = \gamma^0$$

② $i \bar{q}_L^T \not{D} \not{D}^\dagger i G_\mu^a T^{*a} q_L^* \equiv i \bar{q}_L^{d,m} \not{D}^\alpha \not{D}^\beta \not{D}^\gamma \not{D}^\delta i G_\mu^a T^{*a} q_L^{*d,n} :$

$(T_a^{*T})^{nm} = (T_a^\dagger)^{nm} = T^{ann}$ (generators are hermitian)

$= -i \bar{q}_L^{*d,n} \not{D}^\alpha \not{D}^\beta \not{D}^\gamma \not{D}^\delta i G_\mu^a \overbrace{T^{*ann}} q_L^{d,m} =$

$= -i \bar{q}_L^{*d,n} \not{D}^\alpha \not{D}^\beta \not{D}^\gamma \not{D}^\delta i G_\mu^a T^{ann} q_L^{d,m} =$ (omit group indices)

$= -i \bar{q}_L^{*d} \not{D}^\alpha \not{D}^\beta \not{D}^\gamma \not{D}^\delta i G_\mu^a q_L^d :$

$= -i \bar{q}_L^+ \not{D}^\mu \not{D}^\nu i G_\mu^a T^a q_L = -i \bar{q}_L^+ \underbrace{\not{D}^\mu \not{D}^\nu}_{\not{D}^\mu} i G_\mu^a T^a q_L = -i \bar{q}_L^+ \not{D}^\mu i G_\mu^a T^a q_L = -i \bar{q}_L^+ \not{D}^\mu i G_\mu^a T^a q_L$

$\not{D}^\mu = \not{D}^\mu \Rightarrow \not{D}^\mu \not{D}^\nu = \not{D}^\mu$

$\Rightarrow \textcircled{1} + \textcircled{2} = i \bar{q}_L \not{D}^\mu \not{D}^\nu q_L - i \bar{q}_L \not{D}^\mu i G_\mu q_L =$

$= i \bar{q}_L \not{D}^\mu (\underbrace{\not{D}^\nu}_{= D^\mu}) q_L$

kinetic term is correct!

Mass term

suspicion is that it should be q^α & not "-" but "+" in front

$\mathcal{L}_{\text{mass}} = \underbrace{-\frac{1}{2} \bar{q}^\alpha C \varepsilon_{\alpha\beta} \hat{M} q^\beta}_{\textcircled{1}} + \underbrace{\frac{1}{2} \bar{q}^\alpha C \varepsilon_{\alpha\beta} \hat{M}^+ q^\beta}_{\textcircled{2}}$

$\hat{q}_R = \varepsilon \cdot C \bar{q}_L^T \quad (C = i \gamma^2 \gamma^0) \quad \hat{M} = \begin{pmatrix} 0 & 2M \\ M^T & 0 \end{pmatrix}$

① $-\frac{1}{2} (\bar{q}_R, \bar{\tilde{q}}_R) C \varepsilon \hat{M} \hat{q}^T =$

$= -\frac{1}{2} (\bar{q}_R, \bar{\tilde{q}}_R) C \varepsilon \begin{pmatrix} 0 & 2M \\ M^T & 0 \end{pmatrix} \hat{q}^T =$

$= -\frac{1}{2} (\bar{q}_R, \bar{\tilde{q}}_R) C \varepsilon \begin{pmatrix} 0 & 2M \\ M^T & 0 \end{pmatrix} \begin{pmatrix} \bar{q}_R^T \\ \bar{\tilde{q}}_R^T \end{pmatrix} =$

$= -\frac{1}{2} (\bar{q}_R, \bar{\tilde{q}}_R) C \varepsilon \begin{pmatrix} 2M \bar{q}_R^T \\ M^T \bar{\tilde{q}}_R^T \end{pmatrix} =$

$$\overline{q_R} M q_L = \frac{1}{2} \overline{q} C_E M \overline{q}^T$$

$$\mathcal{L} = -\bar{q}_R M q_L + \text{h.c.}$$

$$(-\bar{q}_R M q_L)^+ = \left(-\frac{1}{2} \bar{q} C \varepsilon \hat{M} \bar{q}^T\right)^+ = -\frac{1}{2} \bar{q}^* \hat{M}^+ \varepsilon^+ C^+ \bar{q}^+ \quad \textcircled{=}$$

$$\cdot C = i \gamma^2 \gamma^0 \Rightarrow C^+ = -i \gamma^{0+} \gamma^{2+} = -i \gamma^0 \gamma^0 \gamma^2 \gamma^0 = -i \gamma^2 \gamma^0 = -C$$

$$\textcircled{=} + \frac{1}{2} (\hat{q}^+ \gamma^0)^* \hat{M}^+ \varepsilon^+ C (\hat{q}^+ \gamma^0)^+ = \frac{1}{2} \hat{q}^T \gamma^0 \hat{M}^+ \varepsilon^+ \underbrace{C \gamma^0}_{=-C} \hat{q}^+ =$$

$$= -\frac{1}{2} \hat{q}^T \hat{M}^+ \varepsilon^+ C \hat{q}^+ = -\frac{1}{2} \hat{q}^T \hat{M}^+ \varepsilon^* C \hat{q}^+$$

$$\cdot \gamma^0 C \gamma^0 = \gamma^0 ; \gamma^2 \gamma^0 \gamma^0 = i \gamma^0 \gamma^2 = -i \gamma^2 \gamma^0 = -C$$

Moreover: $(-\bar{q}_R M q_L)^+ = -q_L^+ M^+ (q_R^+ \gamma^0)^+ = -q_L^+ M^+ \gamma^0 q_R = -\bar{q}_L M^+ q_R$

$$\Rightarrow \bar{q}_L M^+ q_R = \frac{1}{2} \hat{q}^T \hat{M}^+ \varepsilon^* C \hat{q}^+$$

Using $\hat{M}^+ = \gamma \hat{M}$ we get

$$\bar{q}_L M^+ q_R = \frac{1}{2} \hat{q}^T \hat{M} \varepsilon^* C \hat{q}^+$$

$\Rightarrow$ Full Lagrangian:

$$\mathcal{L} = \bar{q} i \gamma^\mu D_\mu \hat{q} - \underbrace{\frac{1}{2} \hat{q}^T \hat{M} \varepsilon^* C \hat{q}^+}_{\bar{q}_L M^+ q_R} - \underbrace{\frac{1}{2} \bar{q} C \varepsilon \hat{M} \bar{q}^T}_{\bar{q}_R M q_L}$$

Condensate

$$\langle \bar{q} q \rangle = \sum_{i=1}^{N_f} (\langle \bar{q}_{L,i} q_{R,i} \rangle + \langle \bar{q}_{R,i} q_{L,i} \rangle) = \frac{1}{2} \langle \hat{q}^T \varepsilon^* C \hat{J} \hat{q} + \bar{q} \varepsilon C \hat{J} \bar{q}^T \rangle$$

$$\hat{J} \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

(this is like Poyers's 2.21
 $\chi_c \equiv 2 \langle \bar{q} q \rangle$)

Assume now that the condensate

Transformation

$\hat{q}$ transforms as q_R , i.e.

$$q_R \rightarrow h q_R$$

$$\hat{q} \rightarrow \hat{h} \hat{q}, \quad \hat{h} = \begin{pmatrix} h & 0 \\ 0 & h \end{pmatrix}$$

↑
gauge transformation

Flavor symmetry: $\bar{d}_{kin} \hat{q}^\dagger i \not{D} \hat{q}$

$$\Rightarrow \text{if } \hat{g} \in SU(2N_f) \Rightarrow$$

$$\hat{q} \rightarrow \hat{g} \hat{q}$$

$$\hat{q}^\dagger \rightarrow \hat{q}^\dagger \hat{g}^\dagger$$

$$\Rightarrow d_{kin} = \text{inv}$$

] concatenate: $\hat{q}^\dagger J q \rightarrow \hat{q}^\dagger \hat{g}^\dagger J \hat{g} q = \text{inv if}$

$$\hat{g}^\dagger J \hat{g} = J \Rightarrow \hat{g} \in Sp(2N_f) \subset SU(2N_f)$$

unbroken

For generators Q^a this means:

$$\hat{g} = e^{i\varepsilon^a Q^a}, \quad Q^a \in su(2N_f)$$

$$(1 + i\varepsilon^a Q^a)^T J (1 + i\varepsilon^a Q^a) = J$$

$$\Rightarrow i\varepsilon^a Q^{aT} J + i\varepsilon^a J Q^a = 0$$

$$\Rightarrow Q^{aT} J + J Q^a = 0$$

In case of same masses we expect the condensates for all species to be the same and also $\langle \bar{q}_L q_R \rangle = \langle \bar{q}_R q_L \rangle$.

We have a condensate matrix:

$$(\text{we have } \frac{1}{2} \hat{q}^\dagger J \varepsilon^* C \hat{q} = \bar{q}_L q_R)$$

$$\langle \hat{q}_i^T \varepsilon^* C \hat{q}_i \rangle = \delta J_{ij}, \quad \delta = \frac{\langle \bar{q}_L q_R \rangle}{N_f}$$

$$\Rightarrow J_{ij} \langle \hat{q}_i^T \varepsilon^* C \hat{q}_i \rangle = \delta \underbrace{J_{ij} J_{ij}}_{= \text{Tr}(J J^T) = \text{Tr}(\gamma J J)} =$$

$$= \gamma \text{Tr} \left[\begin{pmatrix} 0 & \gamma \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \gamma \\ 1 & 0 \end{pmatrix} \right] = \gamma \text{Tr} \left[\begin{pmatrix} \gamma & 0 \\ 0 & \gamma \end{pmatrix} \right] =$$

$$= \gamma^2 \text{Tr} \left[\begin{pmatrix} 1_{N_f} & 0 \\ 0 & 1_{N_f} \end{pmatrix} \right] = 2 N_f$$

$$\Rightarrow \langle \hat{q}_i^T \varepsilon^* J_{ij} C \hat{q}_i \rangle = \delta 2 N_f = \frac{\langle \bar{q}_L q_R \rangle}{N_f} \cdot 2 N_f = 2 \langle \bar{q}_L q_R \rangle \quad \checkmark$$

Transformation of cond. under G : $\hat{q}_i \rightarrow g_{im} \hat{q}_m$, $g \in SU(2N_f)$

$$K_{ij} \equiv \langle \hat{q}_i^T \varepsilon^* C \hat{q}_j \rangle \longrightarrow \langle (g_{im} \hat{q}_m)^T \varepsilon^* C g_{jk} \hat{q}_k \rangle = \langle \hat{q}_m^T g_{mi} \varepsilon^* C g_{jk} \hat{q}_k \rangle = g_{mi} g_{jk} K_{mk}$$

$$= g_{mi} K_{mk} g_{kj}^T \quad \text{and} \Rightarrow J_{ij} \rightarrow g J_s g^T$$

In both real & pseudoreal cases the matrix C ensures the antisymmetrization with respect to spin indices to produce a spin singlet. In the real case the matrix ε is symmetric and the condensate is a symmetric product of 2 vector representations. In the pseudoreal case the matrix ε is antisymmetric and the condensate is an antisymmetric product of 2 fundamental representations. To fulfil the Pauli principle the product of the reps. in the real case should be symmetric while in the pseudoreal - antisymmetric

cite kogut for a full discussion

Here goes smth like:

CCWZ construction $\langle \dots \rangle = \delta J_{ij}$

The EFT for the goldstone modes can be written in terms of fluctuations of the orientation of the chiral condensate which

equilibrium value can be chosen as J . Under $SU(2N_f) : J \rightarrow g J g^T$

The Goldstone manifold given by $SU(2N_f)/Sp(2N_f)$ for the pseudoreal case and $SU(2N_f)/SO(2N_f)$ in the pseudoreal case can be parametrized by

using the broken part and the matrix $u \in G/H$ transforms as $U = u J u^T$, where $u = e^{\frac{i}{\sqrt{2}F} \pi^a X^a}$ (X^a -broken)

cite Labosh
cite Tomas
cite CWZ $u \rightarrow g u h^+$
(and $\Rightarrow U \rightarrow g U g^T$).

The covariant derivatives are defined using that fact that

$g (\partial_\mu - i V_\mu) g^+ \in \text{Lie algebra}$ (V_μ -external fields transforming as

$$V_\mu \rightarrow g V_\mu g^+ + i g \partial_\mu g^+)$$

and thus

$$u^+ (\partial_\mu - i V_\mu) u \equiv \Gamma_\mu - \frac{i}{2} u_\mu, \quad \Gamma_\mu = \underset{\substack{\uparrow \\ \text{conserved}}}{T_\mu} Q^a, \quad u_\mu = \underset{\substack{\uparrow \\ \text{broken}}}{u_\mu^a} X^a$$

with transformation rules

$$\Gamma_\mu \rightarrow h \Gamma_\mu h^+ + h \partial_\mu h^+, \quad u_\mu \rightarrow h u_\mu h^+.$$

The external vector field V_μ appears in the field strength

$$V_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu - i [V_\mu, V_\nu].$$

This can be made simpler using

$$f_{\mu\nu} \equiv u^+ V_{\mu\nu} u \rightarrow h f_{\mu\nu} h^+.$$

Not only external vector field can appear but also scalar field $\hat{\chi} \rightarrow g \hat{\chi} g^+$ which can be simplified with

$$\chi \equiv u^+ \hat{\chi} u \rightarrow h \chi h^+$$

If $\exists$ extra discrete symmetries like parity that leave the unbroken part of the group invariant, objects O can be

split into independent parts $O_{\pm} = O \pm P(O)$

To get u_{μ} we can use that fact that a general matrix F can be split into 2 parts, one behaving as the broken part, the other as the unbroken part:

here formulas 3.31 with J instead of J

(also add $u J_s = J_s u^T$, $J_s u = u^T J_s$)

$\Rightarrow$ formula (3.32)

(3.34, 3.35)

The lagrangians are (3.18), (3.4)

External vectors

$$\mathcal{L} = \bar{q}_L \gamma^\mu \ell_{\mu j} q_{Rj} + \bar{q}_R \gamma^\mu \Gamma_{\mu j} q_{Rj}$$

In a new form: $\mathcal{L} = \bar{\hat{q}} \gamma^\mu \hat{V}_\mu \hat{q}$, $\hat{V}_\mu = \begin{pmatrix} \Gamma_\mu & 0 \\ 0 & -\ell_\mu^T \end{pmatrix}$

$$\hat{q} = \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix}, \quad \tilde{q}_R = \varepsilon C \bar{q}_L^T$$

$$\mathcal{L} = \bar{\hat{q}} \gamma^\mu \hat{V}_\mu \hat{q} = \hat{q}^\dagger \gamma^0 \gamma^\mu \hat{V}_\mu \hat{q} = (q_R^\dagger, \tilde{q}_R^\dagger) \gamma^0 \gamma^\mu \hat{V}_\mu \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix} =$$

$$= (q_R^\dagger, (\varepsilon C \bar{q}_L^T)^\dagger) \gamma^0 \gamma^\mu \hat{V}_\mu \begin{pmatrix} q_R \\ \varepsilon C \bar{q}_L^T \end{pmatrix} =$$

$$= (q_R^\dagger, (\varepsilon C \bar{q}_L^T)^\dagger) \gamma^0 \gamma^\mu \begin{pmatrix} \Gamma_\mu & 0 \\ 0 & -\ell_\mu^T \end{pmatrix} \begin{pmatrix} q_R \\ \varepsilon C \bar{q}_L^T \end{pmatrix} =$$

$$= \underbrace{q_R^\dagger \gamma^0 \gamma^\mu \Gamma_\mu q_R}_{= \bar{q}_R \gamma^\mu \Gamma_\mu q_R} - \underbrace{(\varepsilon C \bar{q}_L^T)^\dagger \gamma^0 \gamma^\mu \ell_\mu^T}_{(*)} \cdot \underbrace{\varepsilon C \bar{q}_L^T}_{(**)} \quad \textcircled{=}$$

$$(*) = (\varepsilon C \bar{q}_L^T)^\dagger = (\varepsilon C (q_L^\dagger \gamma^0)^T)^\dagger = (\varepsilon C \gamma^0 q_L^*)^\dagger = q_L^T \gamma^0 C^\dagger \varepsilon^\dagger$$

$$(**) = \bar{q}_L^T = (q_L^\dagger \gamma^0)^T = \gamma^0 q_L^*$$

$$\textcircled{=} \bar{q}_R \gamma^\mu \Gamma_\mu q_R - q_L^T \gamma^0 \underline{C^\dagger} \underline{\varepsilon^\dagger} \gamma^0 \gamma^\mu \ell_\mu^T \cdot \varepsilon \underline{C} \gamma^0 q_L^* =$$

$$= \bar{q}_R \gamma^\mu \Gamma_\mu q_R - q_L^T \gamma^0 \gamma^{\mu*} \underbrace{\varepsilon^\dagger \ell_\mu^T \varepsilon}_{=1} q_L^* = \bar{q}_R \gamma^\mu \Gamma_\mu q_R - q_L^T \gamma^0 \gamma^{\mu*} \ell_\mu^T q_L^*$$

$$\cdot \gamma^0 C^\dagger \gamma^0 \gamma^\mu C \gamma^0 = \gamma^0 (\gamma^2 \gamma^0)^\dagger \gamma^0 \gamma^\mu \gamma^2 \gamma^0 = -i \overbrace{\gamma^0 \gamma^0}^1 \gamma^2 \gamma^0 \gamma^\mu \gamma^2 \gamma^0 =$$

$$= \gamma^2 \gamma^0 \gamma^\mu \gamma^2 = \gamma^0 \gamma^2 \gamma^0 \gamma^\mu \gamma^2 = \gamma^0 \gamma^2 \gamma^\mu \gamma^2 = \gamma^0 \gamma^{\mu*}$$

$$\bullet - q_L^T \gamma^0 \gamma^{\mu*} \ell_{\mu}^T q_L^* = - q_L^{i,d} \gamma^0_{\alpha\beta} \gamma^{\mu*}_{\beta\delta} \ell_{\mu,ik}^T q_{L\delta,k}^* =$$

$$= + q_{L\delta,k}^* \gamma^0_{\alpha\beta} \gamma^{\mu*}_{\beta\delta} \ell_{\mu,ik}^T q_L^{i,d} = q_L^*_{\delta,k} \gamma^{\mu*}_{\beta\delta} \gamma^0_{\alpha\beta} \ell_{\mu,ik}^T q_L^{i,d} =$$

$$= q_{L\delta,k}^* \gamma^{\mu+}_{\sigma\beta} \gamma^0_{\beta\alpha} \ell_{\mu,ki} q_L^{i,d} = q_L^* \gamma^{\mu+} \gamma^0 \ell_{\mu} q_L = q_L^+ \underbrace{\gamma^0 \gamma^0 \gamma^{\mu+} \gamma^0}_{\gamma^{\mu}} \ell_{\mu} q_L =$$

$$= \overline{q_L} \gamma^{\mu} \ell_{\mu} q_L$$

Vector sources

$$\hat{V}_\mu = \begin{pmatrix} r_\mu & 0 \\ 0 & -l_\mu^T \end{pmatrix},$$

$$\mathcal{L} \supset \bar{q} \gamma^\mu \hat{V}_\mu q$$

$$J = \begin{pmatrix} 0 & \gamma \\ 1 & 0 \end{pmatrix}, \quad \gamma^2 = 1$$

(3.33) ?! $\hat{\ell}_\mu = -J \hat{V}^T J^T = - \begin{pmatrix} 0 & \gamma \\ 1 & 0 \end{pmatrix} \begin{pmatrix} r_\mu^T & 0 \\ 0 & -l_\mu \end{pmatrix} \begin{pmatrix} 0 & 1 \\ \gamma & 0 \end{pmatrix} =$

$$= - \begin{pmatrix} 0 & \gamma \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & r_\mu^T \\ -l_\mu \gamma & 0 \end{pmatrix} = - \begin{pmatrix} -l_\mu & 0 \\ 0 & r_\mu^T \end{pmatrix} = \begin{pmatrix} l_\mu & 0 \\ 0 & -r_\mu^T \end{pmatrix}$$

?! $\hat{r}_\mu = \hat{V}_\mu = \begin{pmatrix} r_\mu & 0 \\ 0 & -l_\mu^T \end{pmatrix}$

Conserved current:

$$\hat{V}_\mu = \hat{r}_\mu + \hat{\ell}_\mu = \begin{pmatrix} r_\mu & 0 \\ 0 & -l_\mu^T \end{pmatrix} + \begin{pmatrix} l_\mu & 0 \\ 0 & -r_\mu^T \end{pmatrix} = \begin{pmatrix} r_\mu + l_\mu & 0 \\ 0 & -(r_\mu + l_\mu)^T \end{pmatrix}$$

Broken current:

$$\hat{a}_\mu = \hat{r}_\mu - \hat{\ell}_\mu = \begin{pmatrix} r_\mu - l_\mu & 0 \\ 0 & -(l_\mu - r_\mu)^T \end{pmatrix}$$

rotates left & right fields
in the opposite way

← this current corresponds to
rotating left & right fields
in the same way.

I.e. Broken current is:

$$\hat{a}_\mu = \hat{V}_\mu + J \hat{V}_\mu^T J^T$$

$$\hat{v}_\mu = \hat{V}_\mu - J \hat{V}_\mu^T J^T$$

If we plug $X^a, X^a, a=1, \dots, 5$
then what we get is

that $2\hat{a}_\mu$ corresponds to the broken generators
and $2\hat{v}_\mu$ corresponds to the unbroken generators

Let's pick up just one example: V_μ

$$\mathcal{L} = \bar{q}_{Li} \delta^\mu \ell_{\mu ij} q_{Lj} + \bar{q}_{Ri} \delta^\mu r_{\mu ij} q_{Rj}, \quad i, j = 1, N_F$$

$$P_L q = q_L, \quad P_R q = q_R \quad P_{R/L} = \frac{1 \pm \gamma^5}{2} \quad 1 = P_R + P_L$$

Suppose: $q_i = q_{Li} + q_{Ri} \quad i=1, N_F \quad r_\mu = v_\mu + a_\mu \quad \ell_\mu = v_\mu - a_\mu$

$$\bar{q}_i \delta^\mu v_{\mu ij} q_j + \bar{q}_i \delta^\mu a_{\mu ij} q_j = \bar{q}_i \delta^\mu v_{\mu ij} (q_{Lj} + q_{Rj}) + \bar{q}_i \delta^\mu a_{\mu ij} (-q_{Lj} + q_{Rj}) =$$

$$\gamma^5 P_{R/L} = \gamma^5 \frac{1 \pm \gamma^5}{2} = \frac{\gamma^5 \pm 1}{2} = \frac{\pm 1 + \gamma^5}{2} = \begin{cases} P_R \\ -P_L \end{cases}$$

$$\gamma^5 (P_R + P_L) = -P_L + P_R$$

$$= \bar{q}_{Li} \delta^\mu v_{\mu ij} q_{Lj} + \bar{q}_{Ri} \delta^\mu v_{\mu ij} q_{Rj} - \bar{q}_{Li} \delta^\mu a_{\mu ij} q_{Lj} + \bar{q}_{Ri} \delta^\mu a_{\mu ij} q_{Rj}$$

$$\Rightarrow \bar{q}_{Li} \delta^\mu \underbrace{(v_{\mu ij} - a_{\mu ij})}_{\ell_{\mu ij}} q_{Lj} + \bar{q}_{Ri} \delta^\mu \underbrace{(v_{\mu ij} + a_{\mu ij})}_{r_{\mu ij}} q_{Rj}$$

$$\mathcal{L} = \bar{\hat{q}} i \delta^\mu \overset{\text{gauge}}{D_\mu} \hat{q} - \frac{1}{2} \bar{\hat{q}} \hat{\mathcal{M}} \varepsilon^* C \hat{q} - \frac{1}{2} \bar{\hat{q}} C \varepsilon \hat{\mathcal{M}} \hat{q}^T \quad \hat{\mathcal{M}} = m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = m J$$

$$\hat{q} \equiv \psi \quad - 2N_F \text{ components}$$

$$\Rightarrow \mathcal{L} = i \bar{\psi}_i \delta^\mu D_\mu \psi_j - \frac{1}{2} m \bar{\psi}_i^T J_{ij} \varepsilon^* C \psi_j - \frac{1}{2} m \bar{\psi}_i C \varepsilon J_{ij} \psi_j^T$$

Exercise: apply method of Gell-Mann & Levy to the QCD Lagrangian

$$\mathcal{L}_{\text{QCD}} = i \bar{q}_L \gamma^\mu D_\mu q_L + i \bar{q}_R i D_\mu q_R - \bar{q}_R M q_L + \bar{q}_L M q_R$$

$$] \quad q_L \rightarrow \left(1 - i \sum_a \theta_{La} \frac{\lambda_a}{2}\right) q_L \quad (SU(N_f)_L \text{ transformation})$$

$$\delta \mathcal{L} = \mathcal{L}(\text{new}) - \mathcal{L}(\text{old})$$

$$\theta_{La} - \text{spacetime dependent, real}$$

$$\lambda_a^\dagger = \lambda_a, \quad D_\mu = \partial_\mu + i G_\mu$$

$\Rightarrow$ ① kin term

$$\mathcal{L}(\text{new}) = i \bar{q}_L \left(1 + i \theta_{La} \frac{\lambda_a}{2}\right) \gamma^\mu D_\mu \left[\left(1 - i \theta_{Lb} \frac{\lambda_b}{2}\right) q_L\right] =$$

$$= i \bar{q}_L \left(1 + i \theta_{La} \frac{\lambda_a}{2}\right) \gamma^\mu \left(\partial_\mu q_L - i \frac{\theta_{Lb}}{2} \lambda_b (\partial_\mu q_L)\right) + i \bar{q}_L \left(1 + i \theta_{La} \frac{\lambda_a}{2}\right) \gamma^\mu (i G_\mu) \left(1 - i \theta_{Lb} \frac{\lambda_b}{2}\right) q_L$$

$$= i \bar{q}_L \left(1 + i \theta_{La} \frac{\lambda_a}{2}\right) \gamma^\mu \left(\partial_\mu q_L - i \frac{\theta_{Lb}}{2} \lambda_b \partial_\mu q_L - i \frac{\theta_{Lb}}{2} \lambda_b \partial_\mu q_L\right) + \left(i \bar{q}_L + i \bar{q}_L \theta_{La} \frac{\lambda_a}{2}\right) \gamma^\mu i G_\mu \cdot \left(1 - i \theta_{Lb} \frac{\lambda_b}{2}\right) q_L$$

$$= \left(i \bar{q}_L - \bar{q}_L \theta_{La} \frac{\lambda_a}{2}\right) \gamma^\mu \left(\partial_\mu q_L - i \frac{\theta_{Lb}}{2} \lambda_b \partial_\mu q_L - i \frac{\theta_{Lb}}{2} \lambda_b \partial_\mu q_L\right)$$

$$+ \left(i \bar{q}_L + i \bar{q}_L \theta_{La} \frac{\lambda_a}{2}\right) \gamma^\mu i G_\mu \left(1 - i \theta_{Lb} \frac{\lambda_b}{2}\right) q_L =$$

$$= \underline{i \bar{q}_L \gamma^\mu \left(\partial_\mu q_L - i \frac{\theta_{Lb}}{2} \lambda_b \partial_\mu q_L - i \frac{\theta_{Lb}}{2} \lambda_b \partial_\mu q_L\right)} + \mathcal{O}(\theta^2) +$$

$$\underline{+ i \bar{q}_L \gamma^\mu i G_\mu q_L} + \cancel{i \bar{q}_L \gamma^\mu i G_\mu \left(-i \theta_{Lb} \frac{\lambda_b}{2}\right) q_L} - \cancel{\bar{q}_L \theta_{La} \frac{\lambda_a}{2} \gamma^\mu i G_\mu q_L} + \mathcal{O}(\theta^2)$$

in $\mathcal{L}(\text{old})$

$$\Rightarrow \delta \mathcal{L} =$$

Example: $U(1)$ of Dirac spinor

$$\mathcal{L} = i \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi$$

$$\Rightarrow \text{if } \psi \rightarrow e^{i\varepsilon} \psi = (1+i\varepsilon) \psi$$

$$\Rightarrow \partial_\mu \psi \rightarrow \partial_\mu [(1+i\varepsilon) \psi] = i \partial_\mu \varepsilon \psi + (1+i\varepsilon) \partial_\mu \psi$$

$$\mathcal{L}(\psi + \delta\psi) - \mathcal{L}(\psi) = i \bar{\psi} (1-i\varepsilon) \gamma^\mu (i \partial_\mu \varepsilon \psi + (1+i\varepsilon) \partial_\mu \psi) - m \bar{\psi} (1-i\varepsilon) (1+i\varepsilon) \psi -$$

$$- i \bar{\psi} \gamma^\mu \partial_\mu \psi + m \bar{\psi} \psi = i (\bar{\psi} - i\varepsilon \bar{\psi}) (i \gamma^\mu \partial_\mu \varepsilon \psi + \gamma^\mu (1+i\varepsilon) \partial_\mu \psi) - \cancel{m \bar{\psi} \psi} - \cancel{i\varepsilon m \bar{\psi} \psi} + \cancel{m i\varepsilon \bar{\psi} \psi} + \mathcal{O}(\varepsilon^2)$$

$$- i \bar{\psi} \gamma^\mu \partial_\mu \psi + \cancel{m \bar{\psi} \psi} = i [\bar{\psi} i \gamma^\mu \partial_\mu \varepsilon \psi + \bar{\psi} \gamma^\mu (1+i\varepsilon) \partial_\mu \psi + \cancel{i\varepsilon \bar{\psi} \gamma^\mu \partial_\mu \varepsilon \psi} -$$

$$- i\varepsilon \bar{\psi} \gamma^\mu (1+i\varepsilon) \partial_\mu \psi] - i \bar{\psi} \gamma^\mu \partial_\mu \psi =$$

$$= i [\bar{\psi} i \gamma^\mu \partial_\mu \varepsilon \psi + \cancel{\bar{\psi} \gamma^\mu \partial_\mu \psi} + \cancel{\bar{\psi} \gamma^\mu i\varepsilon \partial_\mu \psi} - \cancel{i\varepsilon \bar{\psi} \gamma^\mu \partial_\mu \psi}] - \cancel{i \bar{\psi} \gamma^\mu \partial_\mu \psi} =$$

$$= - \bar{\psi} \gamma^\mu \psi \partial_\mu \varepsilon \equiv j^\mu \partial_\mu \varepsilon$$

$$\Rightarrow j_\mu = \frac{\partial \mathcal{L}}{\partial (\partial^\mu \varepsilon)} = - \bar{\psi} \gamma_\mu \psi$$

QCD case

$$q_L \rightarrow (1 - i \underbrace{\theta_a \frac{\lambda_a}{2}}_{\delta q_L}) q_L$$

$$D_\mu = \partial_\mu + i G_\mu$$
$$\lambda_a^+ = \lambda_a$$

$$\mathcal{L}_{\text{kin}} = i \bar{q}_L \gamma^\mu D_\mu q_L = i \bar{q}_L \gamma^\mu \partial_\mu q_L - \bar{q}_L \not{G} q_L$$

$$\Rightarrow \delta \mathcal{L}_{\text{kin}} = i \delta \bar{q}_L \gamma^\mu \partial_\mu q_L + i \bar{q}_L \gamma^\mu \partial_\mu \delta q_L - \delta \bar{q}_L \not{G} q_L - \bar{q}_L \not{G} \delta q_L$$

$$= i \bar{q}_L (+i \theta_a \frac{\lambda_a}{2}) \gamma^\mu \partial_\mu q_L + i \bar{q}_L \gamma^\mu \partial_\mu (-i \theta_a \frac{\lambda_a}{2} q_L) - \bar{q}_L (i \theta_a \frac{\lambda_a}{2}) \not{G} q_L -$$

$$- \bar{q}_L \not{G} (-i \theta_a \frac{\lambda_a}{2}) q_L = i \bar{q}_L (i \theta_a \frac{\lambda_a}{2}) \gamma^\mu \partial_\mu q_L - i \bar{q}_L \gamma^\mu i \partial_\mu \theta_a \frac{\lambda_a}{2} q_L$$

$$+ i \bar{q}_L \gamma^\mu (-i) \theta_a \frac{\lambda_a}{2} \partial_\mu q_L = \bar{q}_L \gamma^\mu \partial_\mu \theta_a \frac{\lambda_a}{2} q_L$$

$$\Rightarrow \left\{ j_b^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \theta_b)} = \bar{q}_L \gamma^\mu \frac{\lambda_b}{2} q_L \right\}$$

Now we look at the mass term:

$$\mathcal{L}_{\text{mass}} = -\bar{q}_R \mathcal{M} q_L - \bar{q}_L \mathcal{M} q_R$$

$$\Rightarrow \delta \mathcal{L}_{\text{mass}} = -\bar{q}_R \mathcal{M} \delta q_L - \delta \bar{q}_L \mathcal{M} q_R = -\bar{q}_R \mathcal{M} (-i \theta_a \frac{\lambda_a}{2} q_L) +$$
$$- \bar{q}_L (i \theta_a \frac{\lambda_a}{2}) \mathcal{M} q_R$$

$\Rightarrow$ in total we get

$$\delta \mathcal{L} = \delta \mathcal{L}_{\text{kin}} + \delta \mathcal{L}_{\text{mass}} = \bar{q}_L \gamma^\mu \partial_\mu \theta_a \frac{\lambda_a}{2} q_L + \bar{q}_R \mathcal{M} i \theta_a \frac{\lambda_a}{2} q_L - \bar{q}_L (i \theta_a \frac{\lambda_a}{2}) \mathcal{M} q_R$$

$$\text{in general, } \delta \mathcal{L} = j_a^\mu \partial_\mu \epsilon^a + \epsilon^a \partial_\mu j_a^\mu$$

$$\Rightarrow \partial_\mu j_a^\mu = \frac{\delta \mathcal{L}}{\delta \theta^a} = -(\bar{q}_R \mathcal{M} \frac{\lambda_a}{2} q_L + \bar{q}_L \frac{\lambda_a}{2} \mathcal{M} q_R) \quad \checkmark$$

Now, our case

Gell-Mann - Levi method

$$\mathcal{L} = \bar{\hat{q}} i \gamma^\mu \overset{\text{gauge}}{D}_\mu \hat{q} - \frac{1}{2} \hat{q}^T \hat{\mathcal{M}} \varepsilon^* C \hat{q} - \frac{1}{2} \bar{\hat{q}} C \varepsilon \hat{\mathcal{M}} \hat{q}^T$$

$$\hat{\mathcal{M}} = m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = m J$$

$$D_\mu \hat{q} = \partial_\mu \hat{q} - i G_\mu \hat{q}$$

$$\hat{q} \equiv \psi$$

$$\psi \rightarrow g \psi = (1 + i \varepsilon^a T^a) \psi$$

$$\delta \psi = i \varepsilon^a T^a \psi$$

$$g = g(x)$$

$$g \in SU(2N_f)$$

$$\varepsilon^a = \varepsilon^a(x)$$

Kin term

$$\mathcal{L} = \bar{\psi} i \gamma^\mu \partial_\mu \psi + \bar{\psi} i \gamma^\mu G_\mu \psi$$

$$\delta \mathcal{L} = \delta \bar{\psi} i \gamma^\mu \partial_\mu \psi + \bar{\psi} i \gamma^\mu \partial_\mu \delta \psi + \delta \bar{\psi} i \gamma^\mu G_\mu \psi + \bar{\psi} i \gamma^\mu G_\mu \delta \psi =$$

$$= \bar{\psi} (-i \varepsilon^a T^a) i \gamma^\mu \partial_\mu \psi + \bar{\psi} i \gamma^\mu \partial_\mu (i \varepsilon^a T^a \psi) + \bar{\psi} (-i \varepsilon^a) T^a i \gamma^\mu G_\mu \psi +$$

$$+ \bar{\psi} i \gamma^\mu G_\mu (i \varepsilon^a T^a) G_\mu \psi$$

$$= \bar{\psi} (-i \varepsilon^a T^a) i \gamma^\mu \partial_\mu \psi + \bar{\psi} i \gamma^\mu i \partial_\mu \varepsilon^a T^a \psi + \bar{\psi} i \gamma^\mu i \varepsilon^a T^a \partial_\mu \psi =$$

$$= - \bar{\psi} i \gamma^\mu \partial_\mu \varepsilon^a T^a \psi$$

mass term

$$- \mathcal{d}_m = \frac{1}{2} \hat{q}^T \hat{\mathcal{M}} \varepsilon^* C \hat{q} + \frac{1}{2} \bar{\hat{q}} C \varepsilon \hat{\mathcal{M}} \hat{q}^T \equiv$$

(ε -color space
C-Dirac space)

$$\equiv \frac{1}{2} \psi^T \hat{\mathcal{M}} \varepsilon^* C \psi + \frac{1}{2} \bar{\psi} C \varepsilon \hat{\mathcal{M}} \bar{\psi}^T$$

$$\bar{\psi} = \psi^\dagger \gamma_0$$

$$T^\dagger = T^T$$

$$\psi \rightarrow g \psi = (1 + i \varepsilon^a T^a) \psi$$

$$\delta \psi = i \varepsilon^a T^a \psi \Rightarrow \delta \bar{\psi} = \bar{\psi} (-i \varepsilon^a) T^a$$

$$\delta \bar{\psi}^T = T^{aT} (-i \varepsilon^a) \bar{\psi}^T$$

$$\delta \psi^T = \psi^T i \varepsilon^a T^T$$

$$\Rightarrow - \delta \mathcal{d}_m = \frac{1}{2} \delta \psi^T \hat{\mathcal{M}} \varepsilon^* C \psi + \frac{1}{2} \psi^T \hat{\mathcal{M}} \varepsilon^* C \delta \psi + \frac{1}{2} \delta \bar{\psi} C \varepsilon \hat{\mathcal{M}} \bar{\psi}^T$$

$$+ \frac{1}{2} \bar{\psi} C \varepsilon \hat{\mathcal{M}} \delta \bar{\psi}^T =$$

$$\begin{aligned}
&= \frac{1}{2} \psi^T i \varepsilon^a T^a \hat{M} \varepsilon^* C \psi + \frac{1}{2} \psi^T \hat{M} \varepsilon^* C i \varepsilon^a T^a \psi + \\
&+ \frac{1}{2} \bar{\psi} T^a (-i \varepsilon^a) C \varepsilon \hat{M} \bar{\psi}^T + \frac{1}{2} \bar{\psi} C \varepsilon \hat{M} T^{aT} (-i \varepsilon^a) \bar{\psi}^T \\
&= \frac{1}{2} \psi^T \left(i \varepsilon^a T^{aT} \hat{M} \varepsilon^* C + \hat{M} \varepsilon^* C i \varepsilon^a T^a \right) \psi + \\
&+ \frac{1}{2} \bar{\psi} \left(T^a (-i \varepsilon^a) C \varepsilon \hat{M} + C \varepsilon \hat{M} T^{aT} (-i \varepsilon^a) \right) \bar{\psi}^T \\
&= \frac{1}{2} i \varepsilon^a \psi^T \left(T^{aT} \hat{M} + \hat{M} T^a \right) \varepsilon^* C \psi - i \varepsilon^a \frac{1}{2} \bar{\psi} \left(T^a \hat{M} + \hat{M} T^{aT} \right) C \varepsilon \bar{\psi}^T
\end{aligned}$$

$\Rightarrow$ totally:

$$\begin{aligned}
\delta \mathcal{L} = & - \bar{\psi} \gamma^\mu \partial_\mu \varepsilon^a T^a \psi + i \varepsilon^a \frac{1}{2} \bar{\psi} \left(T^a \hat{M} + \hat{M} T^{aT} \right) C \varepsilon \bar{\psi}^T \\
& - \frac{1}{2} i \varepsilon^a \psi^T \left(T^{aT} \hat{M} + \hat{M} T^a \right) \varepsilon^* C \psi
\end{aligned}$$

$$\equiv j_a^\mu \partial_\mu \varepsilon^a + \varepsilon^a \partial_\mu j_a^\mu$$

$\Rightarrow$

$$j_a^\mu = - \bar{\psi} \gamma^\mu T^a \psi$$

$$\partial_\mu j_a^\mu = \frac{i}{2} \left[\bar{\psi} \left(T^a \hat{M} + \hat{M} T^{aT} \right) C \varepsilon \bar{\psi}^T - \psi^T \left(T^{aT} \hat{M} + \hat{M} T^a \right) \varepsilon^* C \psi \right]$$

] mass degenerate case : $\mathcal{M} = \underbrace{\begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}}_{N \times N} \} N_\ell = m \cdot 1$

$$\Rightarrow \hat{M} = \begin{pmatrix} 0 & \gamma \mathcal{M} \\ \mathcal{M}^\dagger & 0 \end{pmatrix} = m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = m J$$

$\Rightarrow$ we have conserved current if $T^{aT} J + J T^a = 0$

Note

$$T^a J + J T^{aT} = 0$$

$$\hookrightarrow T^{aT} J + J T^a = 0$$

$\Downarrow$

$$J T^a J^2 + J^2 T^{aT} J = 0$$

$$J^2 = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2$$

$$\Rightarrow J T^a + T^{aT} J = 0$$

same

In the generating functional, we want to add a vector field which will be coupled to the current, such that we can generate corr. functions with the current:

$$j_\mu^a = \bar{\psi} \gamma_\mu T^a \psi$$

$$\Rightarrow \mathcal{L}_{uv} \supset \sum_a V_\mu^a \bar{\psi} \gamma_\mu T^a \psi \equiv \bar{\psi} \gamma_\mu V_\mu \psi, \quad V_\mu^+ = V_\mu$$

in $Sp(4)$ there will be 5 broken & 10 unbroken currents

We don't need h.c. here because this term is hermitian:

$$(\bar{\psi} \gamma_\mu V_\mu \psi)^+ = \psi^\dagger V_\mu^\dagger \gamma_\mu^\dagger \bar{\psi}^\dagger = \psi^\dagger V_\mu^\dagger \gamma_\mu^\dagger (\psi^\dagger \gamma_0)^+ =$$

$$= \psi^\dagger V_\mu \underbrace{\gamma_0 \gamma_\mu \gamma_0}_{=1} \gamma_0^\dagger \psi = \bar{\psi} V_\mu \gamma_\mu \psi = \bar{\psi} \gamma_\mu V_\mu \psi$$

In a new form: $\mathcal{L} = \bar{\hat{q}} \gamma^\mu \hat{V}_\mu \hat{q}$, $\hat{V}_\mu = \begin{pmatrix} \Gamma_\mu & O_\mu \\ O_\mu^+ & -\ell_\mu^T \end{pmatrix}$

$$\hat{q} = \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix}, \quad \tilde{q}_R = \varepsilon C \bar{q}_L^T$$

$$\mathcal{L} = \bar{\hat{q}} \gamma^\mu \hat{V}_\mu \hat{q} = \hat{q}^\dagger \gamma^0 \gamma^\mu \hat{V}_\mu \hat{q} = (q_R^\dagger, \tilde{q}_R^\dagger) \gamma^0 \gamma^\mu \hat{V}_\mu \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix} =$$

$$= (q_R^\dagger, (\varepsilon C \bar{q}_L^T)^+) \gamma^0 \gamma^\mu \hat{V}_\mu \begin{pmatrix} q_R \\ \varepsilon C \bar{q}_L^T \end{pmatrix} =$$

$$= (q_R^\dagger, (\varepsilon C \bar{q}_L^T)^+) \gamma^0 \gamma^\mu \begin{pmatrix} \Gamma_\mu & O_\mu \\ O_\mu^+ & -\ell_\mu^T \end{pmatrix} \begin{pmatrix} q_R \\ \varepsilon C \bar{q}_L^T \end{pmatrix} =$$

$$= (q_R^\dagger, (\varepsilon C \bar{q}_L^T)^+) \gamma^0 \gamma^\mu \begin{pmatrix} \Gamma_\mu q_R + O_\mu \varepsilon C \bar{q}_L^T \\ O_\mu^+ q_R - \ell_\mu^T \varepsilon C \bar{q}_L^T \end{pmatrix} =$$

$$= \underbrace{q_R^\dagger \gamma^0 \gamma^\mu \Gamma_\mu q_R}_{\bar{q}_R \not{V} q_R} + \underbrace{q_R^\dagger \gamma^0 \gamma^\mu O_\mu \varepsilon C \bar{q}_L^T}_{(*)} + (\varepsilon C \bar{q}_L^T)^+ \gamma^0 \gamma^\mu O_\mu^+ q_R - (\varepsilon C \bar{q}_L^T)^+ \gamma^0 \gamma^\mu \ell_\mu^T \varepsilon C \bar{q}_L^T$$

$$(*) \quad \bar{q}_R \gamma^\mu O_\mu \varepsilon C \bar{q}_L^T = \bar{q}_R \gamma^\mu O_\mu \varepsilon \overset{\gamma^0 \gamma^\mu}{C} \gamma^0 q_L^* = \bar{q}_R \gamma^\mu O_\mu \varepsilon \cdot \overset{N_f \times N_f \text{ matrix}}{i \gamma^2 \gamma^0} q_L^* =$$

$$= q_R^\dagger \gamma^0 \gamma^\mu O_\mu \varepsilon \cdot i \gamma^2 q_L^*$$

$$\gamma^0 \gamma^\mu \gamma^2 \quad (\equiv)$$

$$\stackrel{\mu=0}{\equiv} \gamma^2 \gamma^2 = \gamma^2 = \begin{pmatrix} 0 & b_2 \\ -b_2 & 0 \end{pmatrix} \Rightarrow (0, q_R^\dagger) \begin{pmatrix} 0 & b_2 \\ -b_2 & 0 \end{pmatrix} \begin{pmatrix} q_L^* \\ 0 \end{pmatrix} =$$

$$= (0, q_R^\dagger) \begin{pmatrix} 0 \\ -b_2 q_L^* \end{pmatrix} = -q_R^\dagger b_2 q_L^*$$

The condensate @ LO (for the paper)

$$\alpha_{LO} = \frac{F^2}{4} \langle u_\mu u^\mu + \psi_+ \rangle$$

$$\psi_+ = u^\dagger \chi J^T u^\dagger + u J \psi^\dagger u$$

$$J \quad \bar{u} = 0 \Rightarrow u = 1 \Rightarrow \psi_+ = \chi J^T + J \psi^\dagger$$

$$\Rightarrow V_{eff} = - \frac{F^2}{4} \langle \chi J^T + J \psi^\dagger \rangle \quad \textcircled{=}$$

$$\psi = 2B_0 \hat{u} \quad \text{with} \quad \hat{u} = \begin{pmatrix} 0 & \gamma u \\ u^\dagger & 0 \end{pmatrix} \Rightarrow \psi^\dagger = 2B_0 \hat{u}^\dagger = 2B_0 \begin{pmatrix} 0 & (u^\dagger)^\dagger \\ \gamma u^\dagger & 0 \end{pmatrix}$$

$$\begin{matrix} u^\dagger = u \\ \end{matrix} \quad = 2B_0 \begin{pmatrix} 0 & u \\ \gamma u & 0 \end{pmatrix} \quad J = \begin{pmatrix} 0 & \gamma \\ 1 & 0 \end{pmatrix}$$

$$\textcircled{=} - \frac{F^2}{4} \langle 2B_0 \begin{pmatrix} 0 & \gamma u \\ u & 0 \end{pmatrix} J^T + J \cdot 2B_0 \begin{pmatrix} 0 & u \\ \gamma u & 0 \end{pmatrix} \rangle =$$

$$= - \frac{F^2}{4} \cdot 2B_0 \left\langle \begin{pmatrix} 0 & \gamma u \\ u & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ \gamma & 0 \end{pmatrix} + \begin{pmatrix} 0 & \gamma \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & u \\ \gamma u & 0 \end{pmatrix} \right\rangle$$

$$\stackrel{\gamma^2=1}{=} - \frac{F^2 B_0}{2} \left\langle \begin{pmatrix} u & 0 \\ 0 & u \end{pmatrix} + \begin{pmatrix} u & 0 \\ 0 & u \end{pmatrix} \right\rangle =$$

$$= - \frac{F^2 B_0}{2} \cdot 2 \left\langle \begin{pmatrix} u & 0 \\ 0 & u \end{pmatrix} \right\rangle = - F^2 B_0 \left\langle \begin{pmatrix} m_u & & & \\ & m_d & & \\ & & m_u & \\ & & & m_d \end{pmatrix} \right\rangle$$

$$= -2 F^2 B_0 (m_u + m_d)$$

$$V_{eff} = -2 F^2 B_0 (m_u + m_d)$$

$$\Rightarrow \langle \bar{u} u \rangle = \left. \frac{\partial V}{\partial m_u} \right|_{\bar{u}=0} = -2 F^2 B_0$$

$$\langle \bar{d} d \rangle = \left. \frac{\partial V}{\partial m_d} \right|_{\bar{d}=0} = -2 F^2 B_0$$

$$\Rightarrow \langle \bar{q} q \rangle = \langle \bar{u} u \rangle + \langle \bar{d} d \rangle = -4 F^2 B_0$$

but not $-2 F^2 B_0$ like in
Bjerrum.

Go back to the vector currents

We are not sure what to write for the vectr. Let's return to the EFT logic. We just need all the operators we can possibly write.

$$V_\mu \xrightarrow{g} g V_\mu g^\dagger + i g \partial_\mu g^\dagger$$

$$u \xrightarrow{G} h u h^\dagger$$

$$u \rightarrow g u h^\dagger$$

} transf. rules

$$V_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu - i[V_\mu, V_\nu]$$

$$f_{\mu\nu} = u^\dagger V_{\mu\nu} u$$

} definition

just standard field strength

$$V_{\mu\nu} \xrightarrow{g} g V_{\mu\nu} g^\dagger$$

$$\Rightarrow f_{\mu\nu} = u^\dagger V_{\mu\nu} u \xrightarrow{G} (h u^\dagger g^\dagger) (g V_{\mu\nu} g^\dagger) (g u h^\dagger) = h (u^\dagger V_{\mu\nu} u) h^\dagger = h f_{\mu\nu} h^\dagger$$

$$(\langle f_{\mu\nu}^2 \rangle \rightarrow \langle h f_{\mu\nu} h^\dagger h f_{\mu\nu} h^\dagger \rangle = \text{inv})$$

$$f_{\pm\mu\nu} = J u V_{\mu\nu} u^\dagger J^T \pm u V_{\mu\nu} u^\dagger$$

why not $u^\dagger V_{\mu\nu} u$? $u V_{\mu\nu} u^\dagger$ transforms ugly!

$$\langle J_A u^\dagger V_{\mu\nu} u J_A^T u^\mu u^\nu \rangle \rightarrow 2 J_A \cdot h u^\dagger V_{\mu\nu} u h^\dagger J_A^T \cdot h u^\mu \underbrace{h^\dagger h}_1 u^\nu h^\dagger$$

$$= \langle \underbrace{(h^\dagger J_A h)}_{h^* J_A} u^\dagger V_{\mu\nu} u h^\dagger \underbrace{J_A^T h}_{h^* J_A^T} u^\mu u^\nu \rangle$$

$$J_A^T h = \int J_A h = \int J_A (1 + i a \cdot Q) = \int (J_A - i a Q^T J_A) = \int (1 - i a Q^T) J_A =$$

$$= \int e^{-i a Q^T} J_A = \int e^{-i a (Q^T)^{**}} J_A = \int e^{(i a Q^T)^*} J_A =$$

$$= \int h^* J_A = h^* J_A^T$$

$$u^\dagger J_A = (1 + i X^\dagger) J = (J + i X^T J) = (J + i J X) = J u$$



